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(54) **THERMAL STORAGE AND SUPPLY**

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(71) Applicant: **Spirax-Sarco Limited**, Cheltenham (GB)

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(72) Inventors: **Jeremy Miller**, Cheltenham (GB); **Sean Spencer**, Cheltenham (GB); **Kevin Rushbrooke**, Cheltenham (GB); **George Tebbutt**, Bristol (GB)

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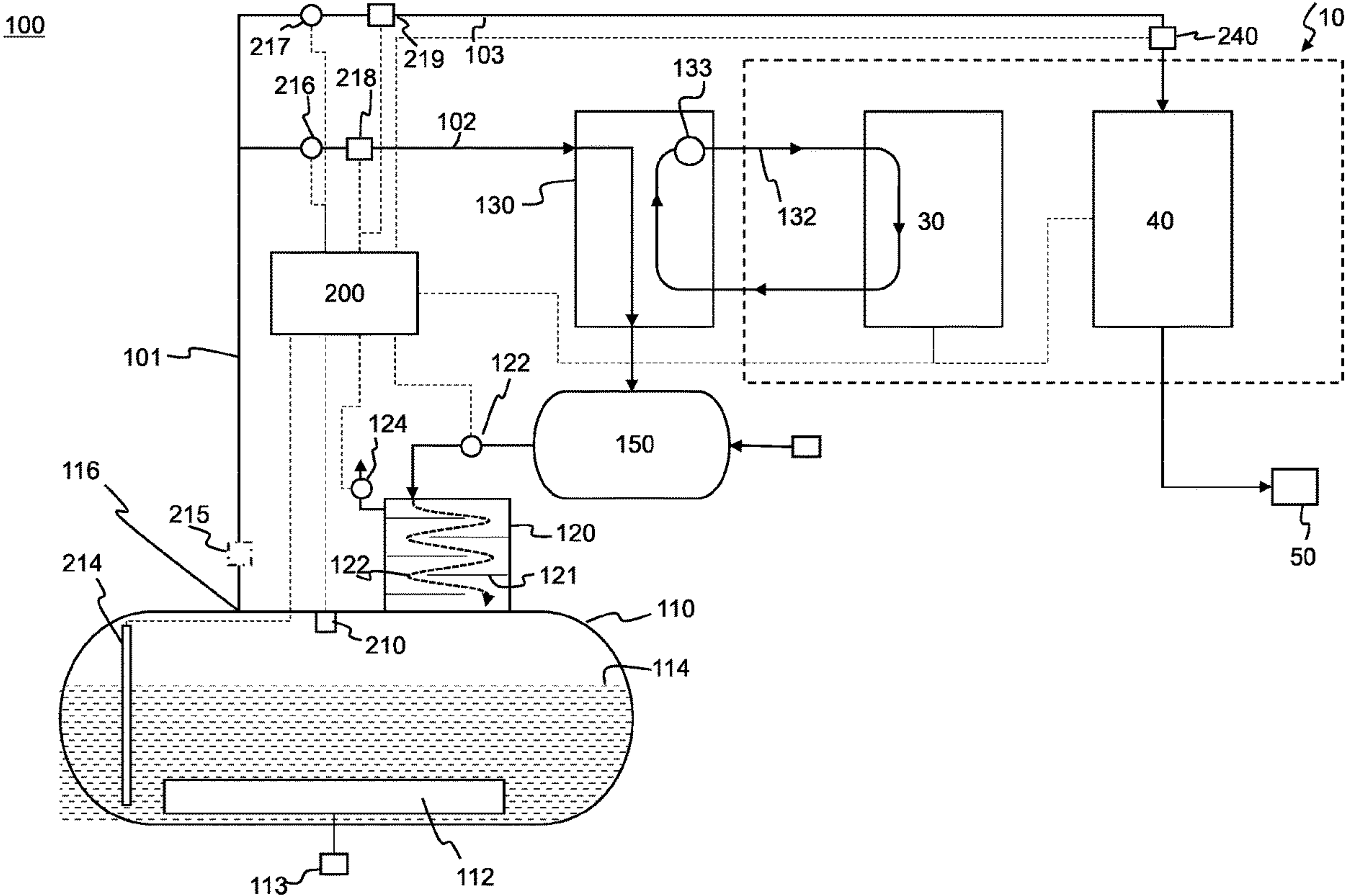
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(57) **ABSTRACT**

There is disclosed a thermal energy storage and supply method that includes providing subcooled water to a pressure vessel; heating liquid water within the pressure vessel using an electrically-powered heater so that the vessel contains saturated liquid water and steam at a variable storage pressure; controlling the heater to raise the storage pressure to a peak storage pressure of at least 2 MPa; and selectively discharging steam from an outlet of the pressure vessel to a thermal load, in response to a thermal energy demand, such that during a depletion period the storage pressure reduces by at least 1 MPa from the peak storage pressure.



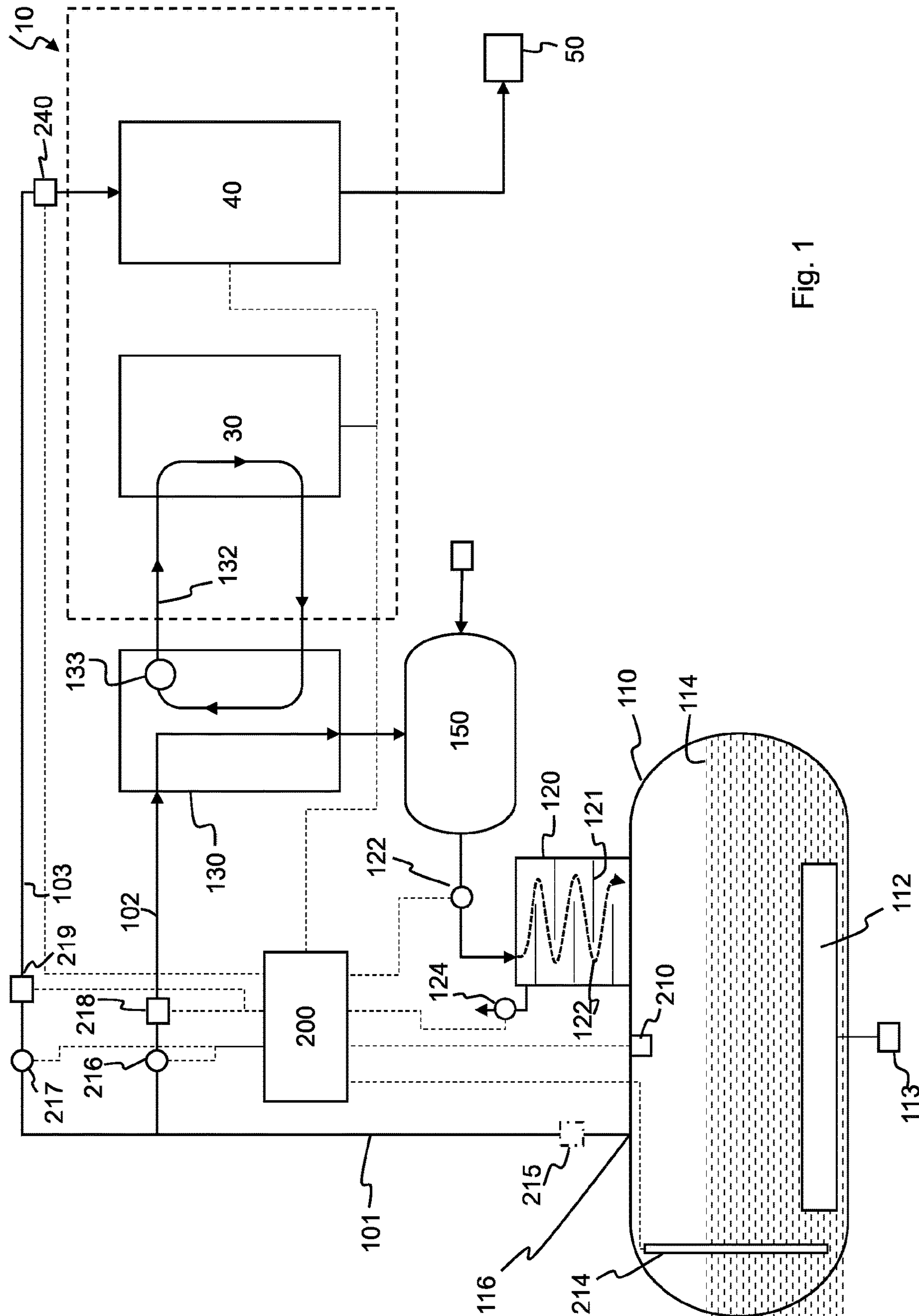


Fig. 1

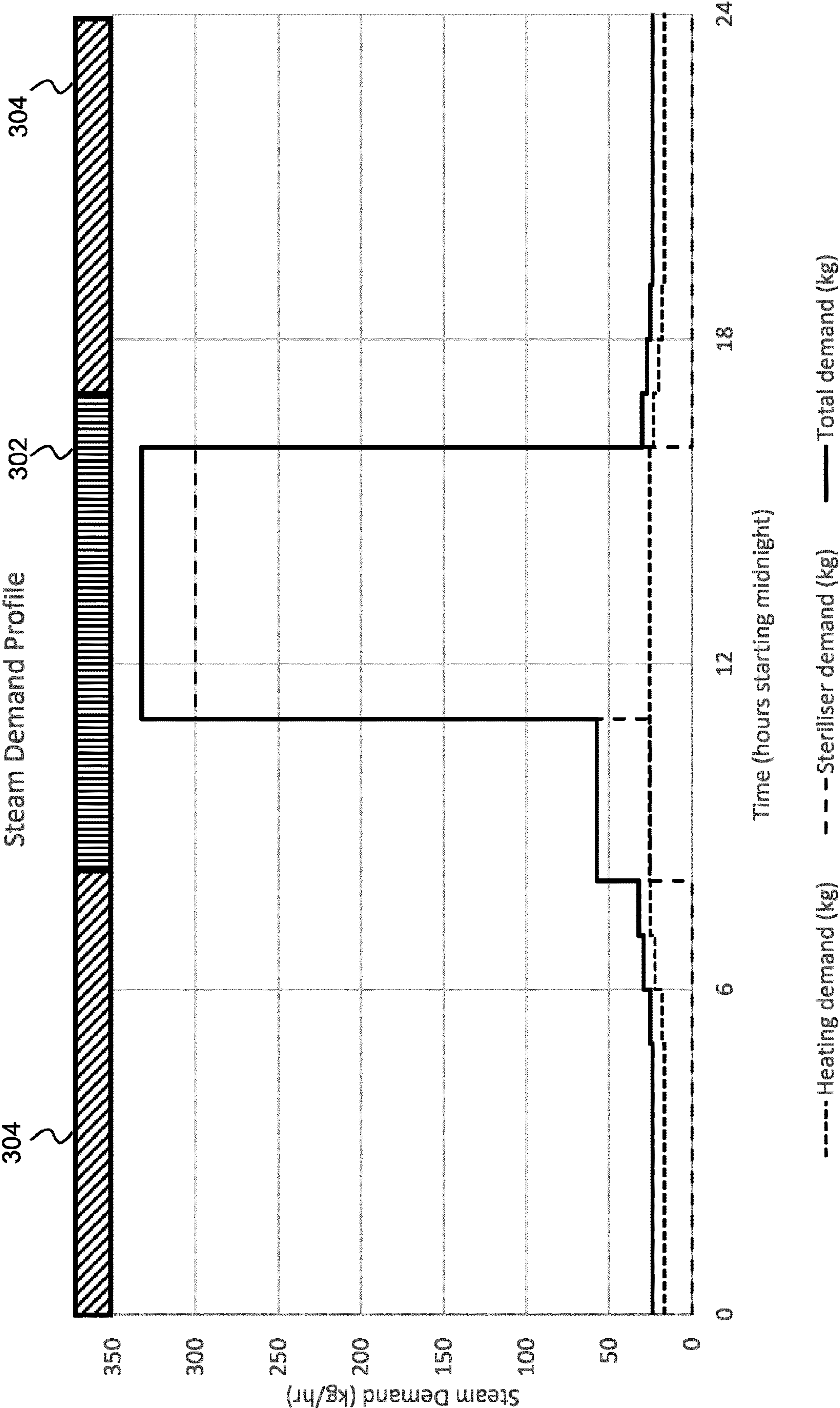


Fig. 2

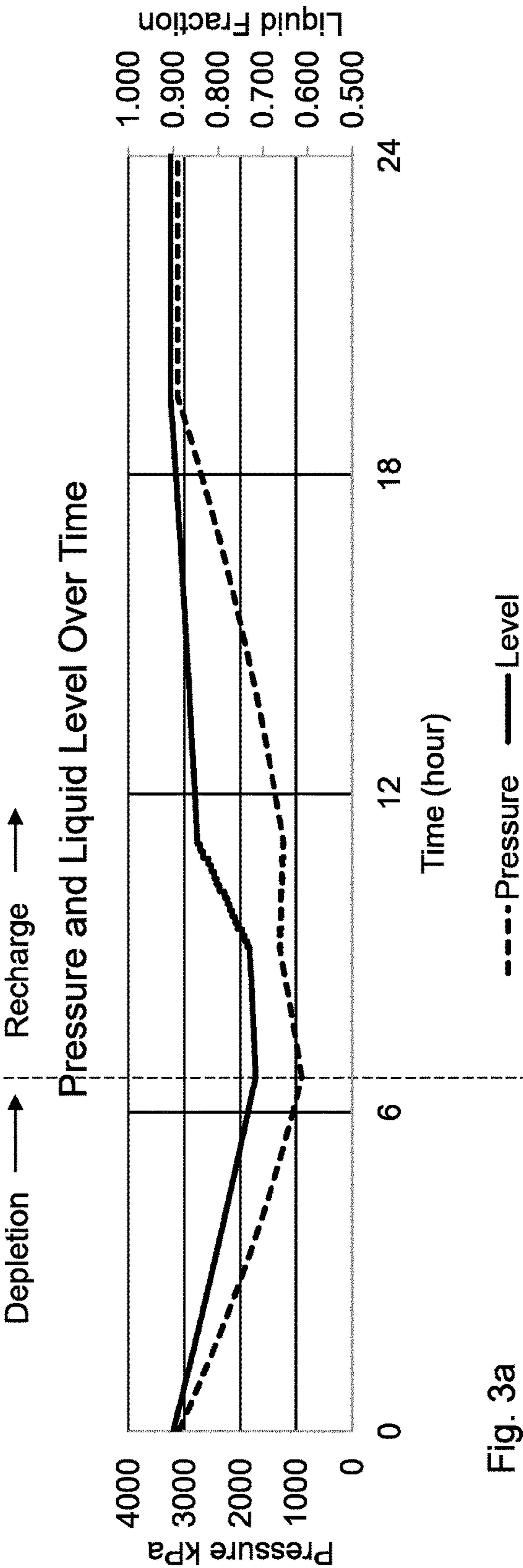


Fig. 3a

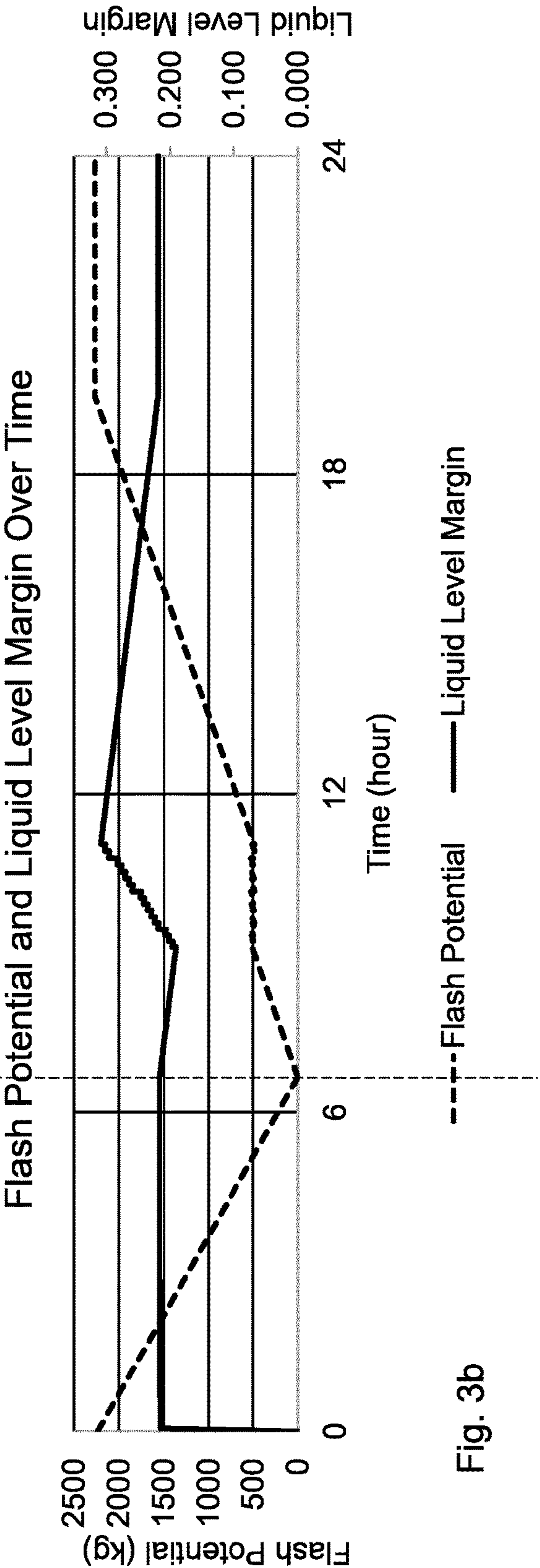


Fig. 3b

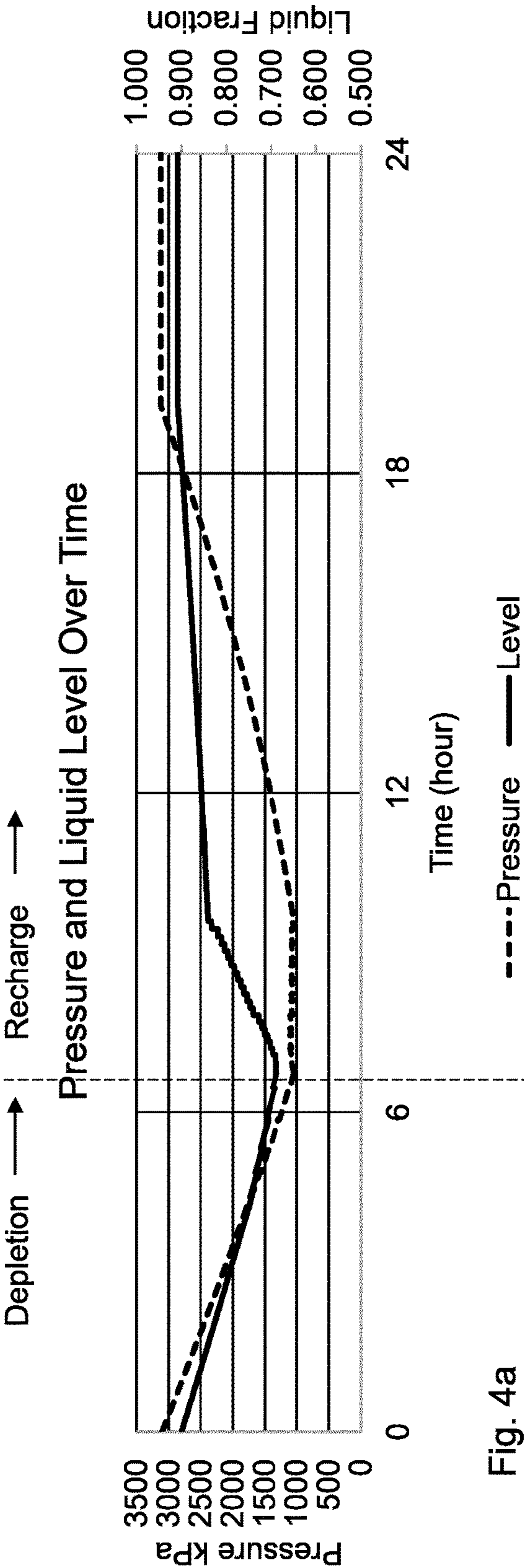


Fig. 4a

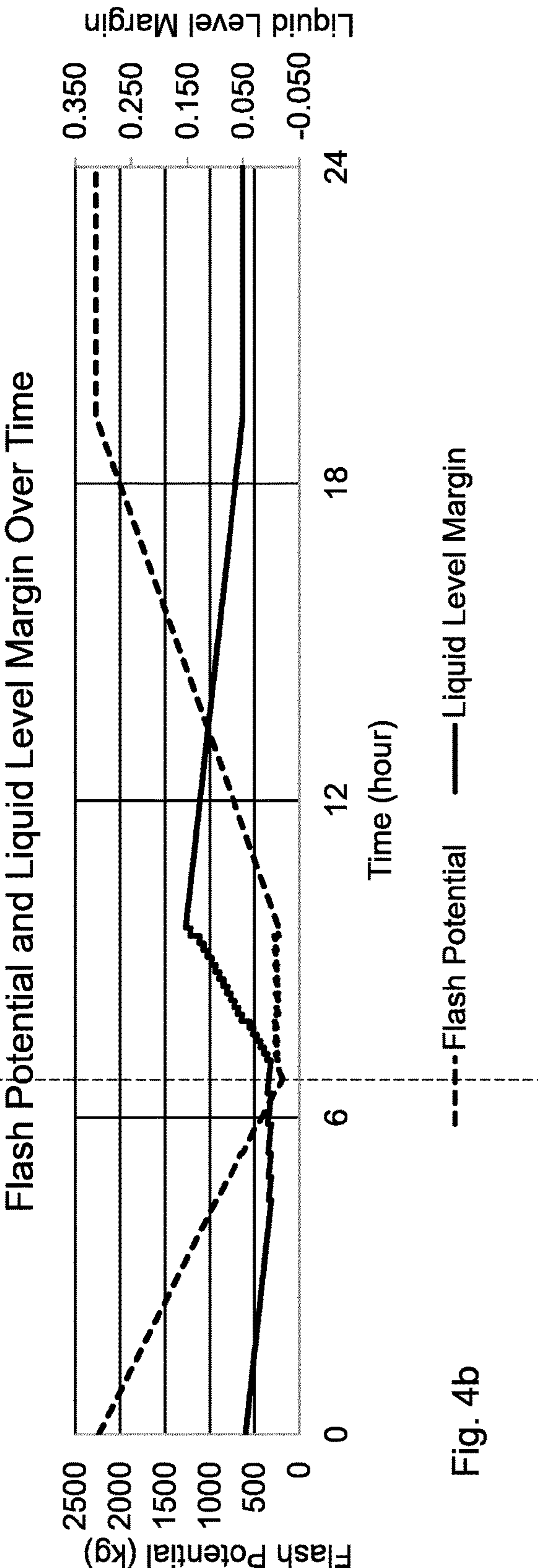
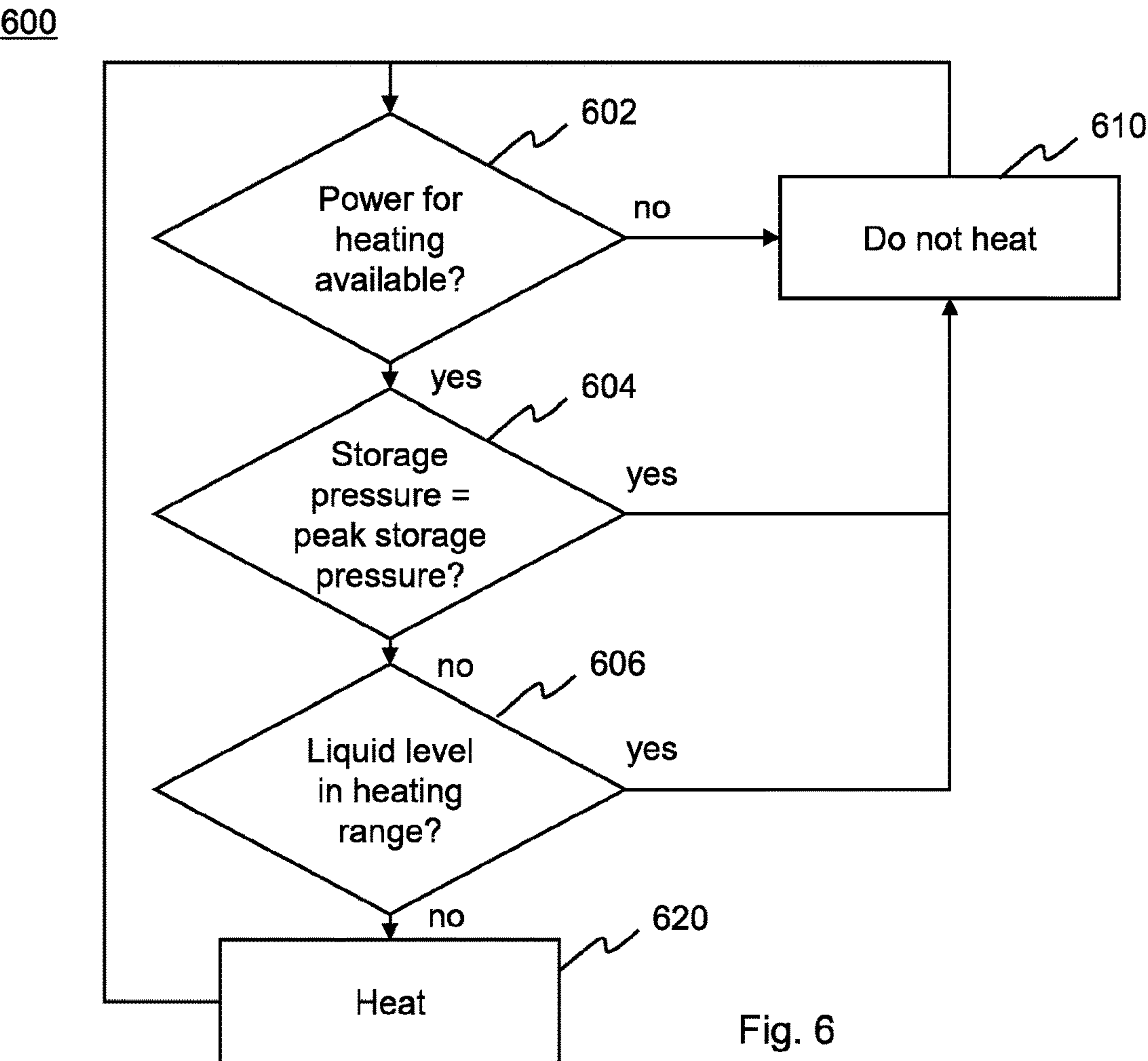
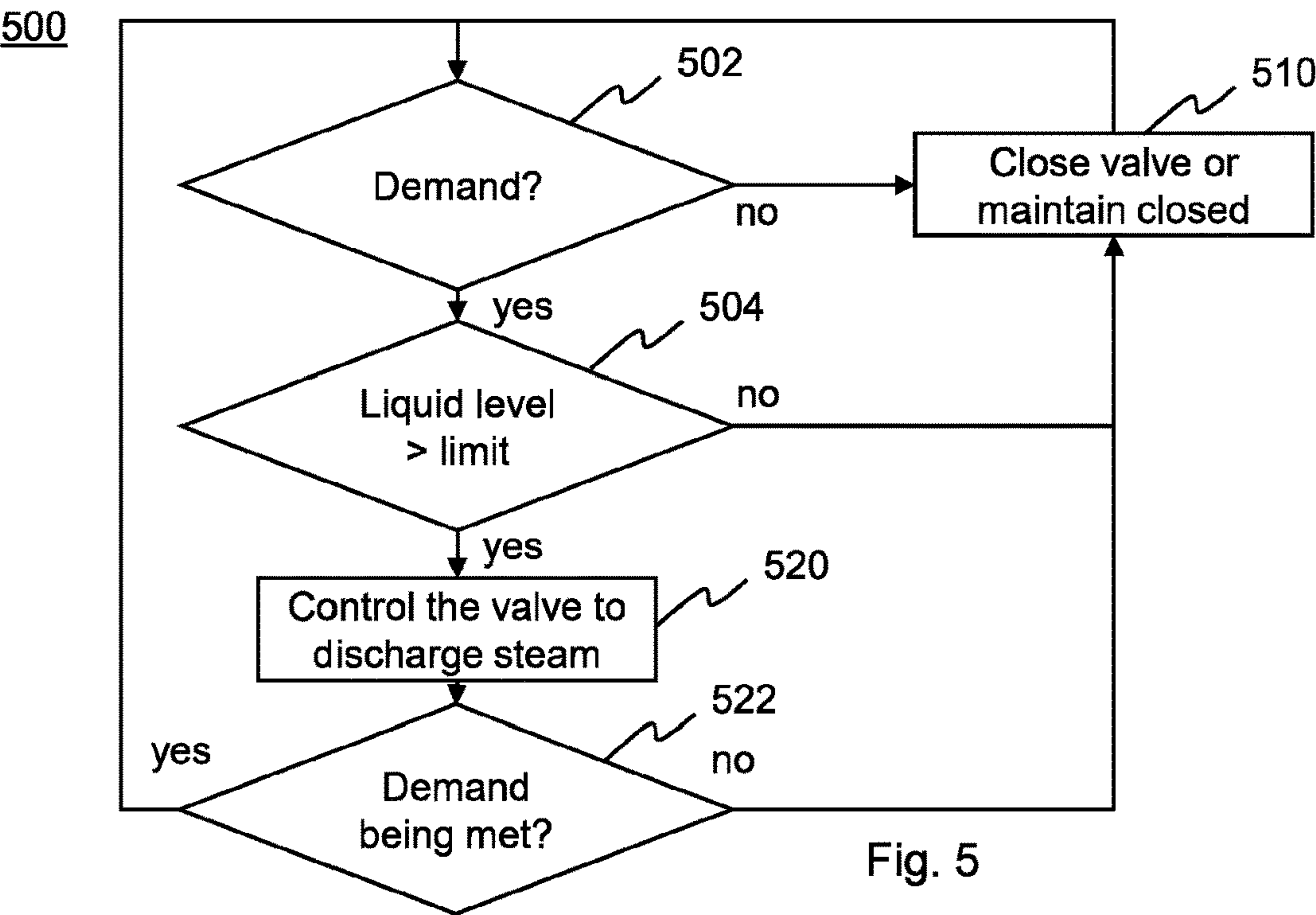


Fig. 4b



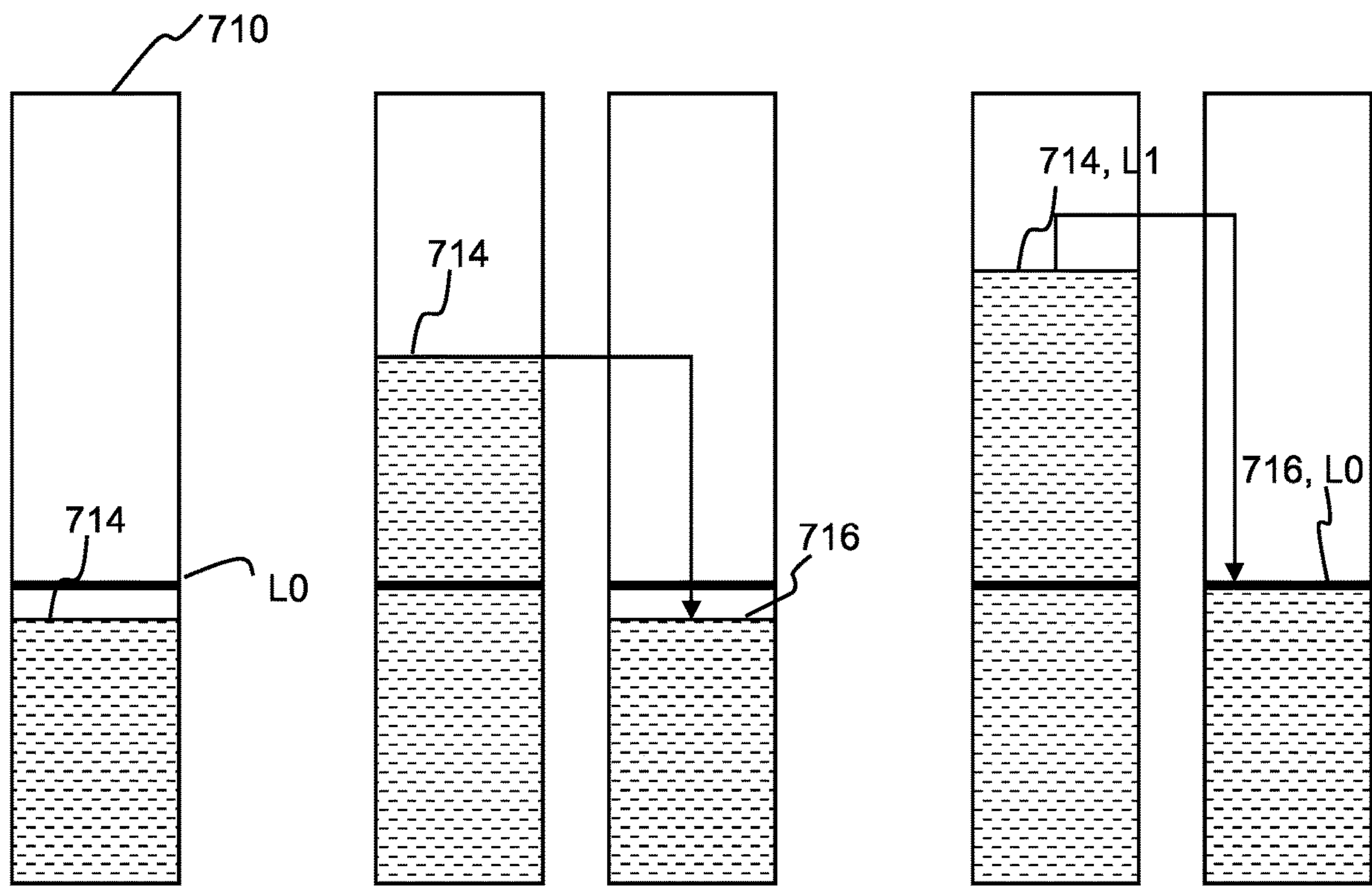


Fig. 7a

Fig. 7b

Fig. 7c

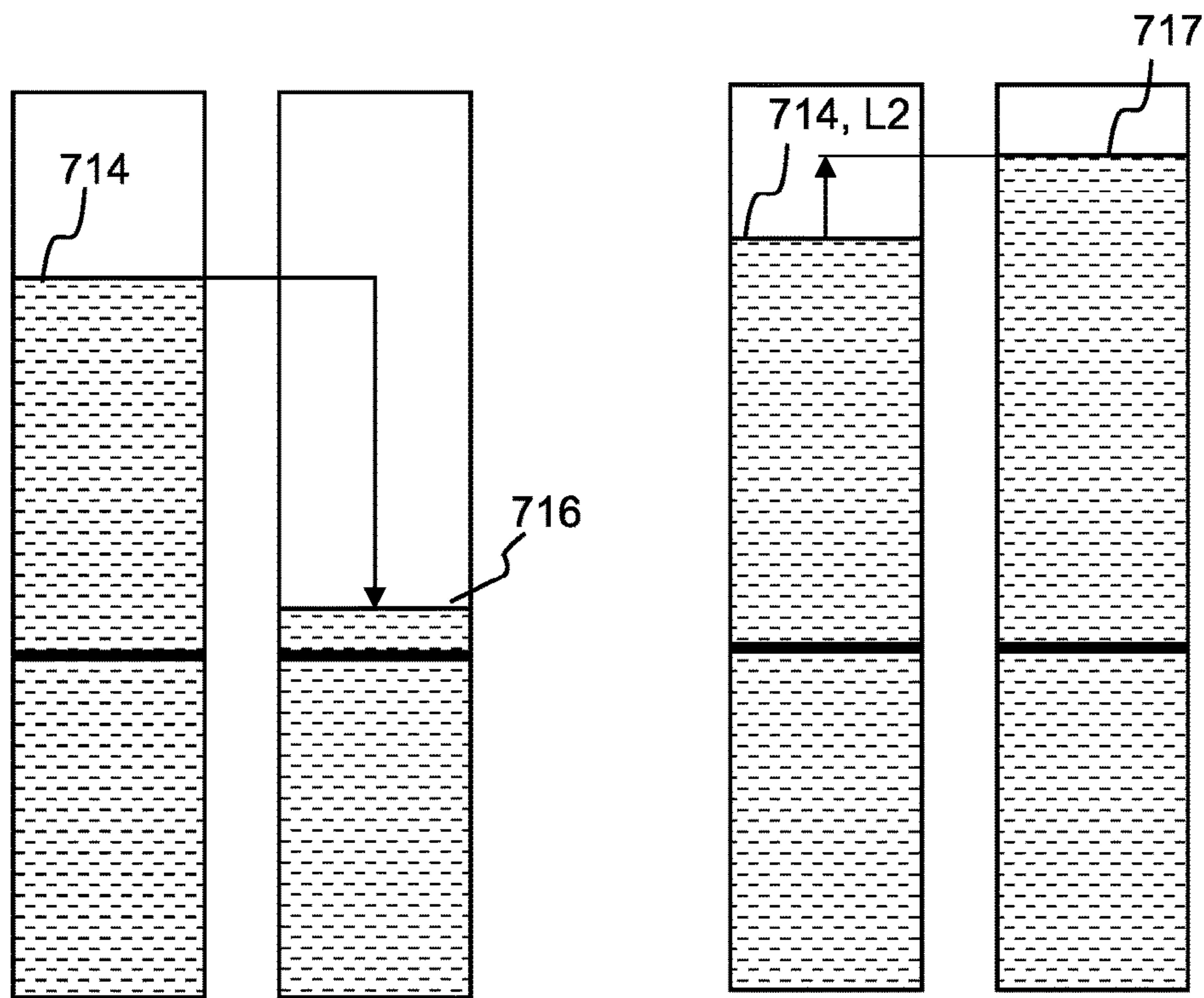
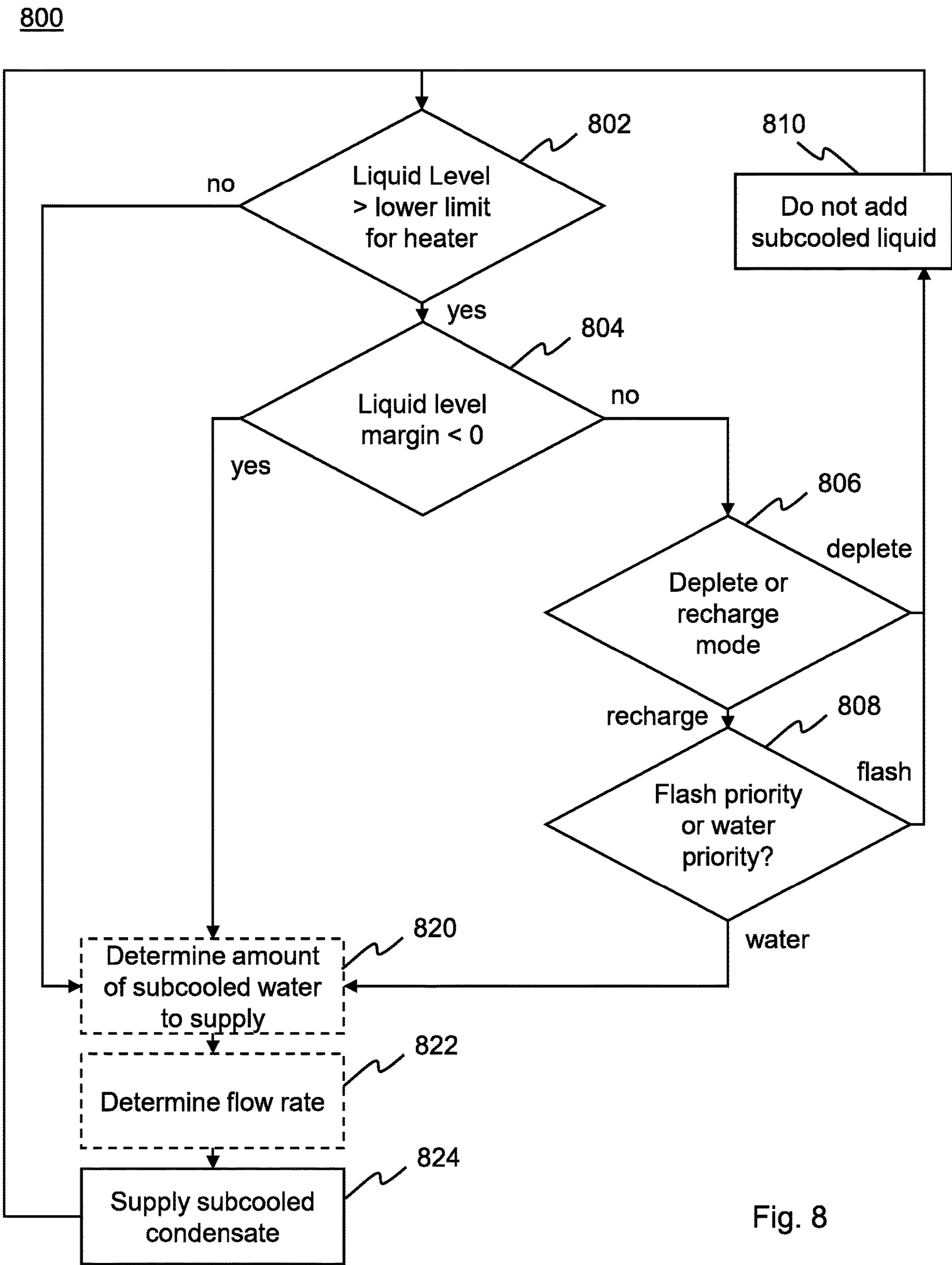


Fig. 7d

Fig. 7e



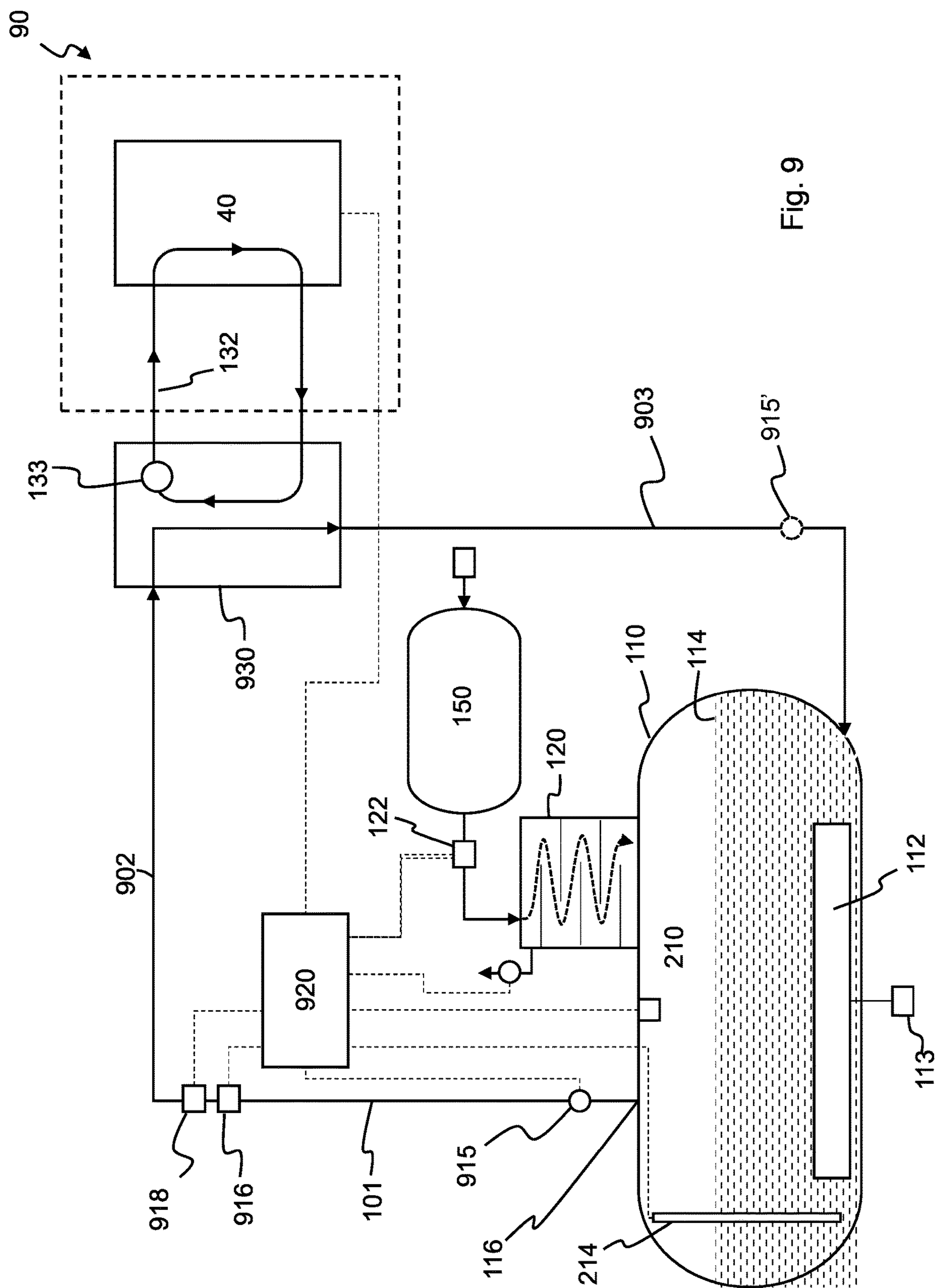


Fig. 9

THERMAL STORAGE AND SUPPLY

[0001] The invention relates to a method and apparatus (e.g. an installation) for thermal energy storage and supply.

BACKGROUND

[0002] It is known to provide industrial plants and other facilities which require thermal energy with a steam supply system capable of generating steam for distribution to thermal loads in the plant or other installation.

[0003] Such steam supply systems typically comprise a burner-powered boiler and an accumulator. It is inefficient to operate a boiler over a large turndown ratio to accommodate variations in steam demand, and so an accumulator is typically provided to accommodate such variations in demand over and under an optimal steam discharge rate of the boiler, during an operational period of the boiler.

[0004] There is a trend to replace systems that require fossil fuels with electrically-powered systems. Electrically-powered boilers with an electrical heating element have been proposed, the electrical heating elements offering more consistent efficiency with a variable power output, such that a variable steam demand can be more efficiently accommodated by the boiler.

SUMMARY

[0005] According to a first aspect there is disclosed a method of thermal energy storage and supply, comprising:

[0006] providing subcooled water to a pressure vessel;

[0007] heating liquid water within the pressure vessel using an electrically-powered heater so that the vessel contains saturated liquid water and steam at a variable storage pressure;

[0008] controlling the heater to raise the storage pressure to a peak storage pressure of at least 2 MPa; and

[0009] selectively discharging steam from an outlet of the pressure vessel to a thermal load, in response to a thermal energy demand, such that during a depletion period the storage pressure reduces by at least 1 MPa from the peak storage pressure.

[0010] Numerical pressure values disclosed herein are absolute pressures except where indicated (e.g. as a gauge pressure). The depletion period as referred to herein may be a period in which an enthalpy output corresponding to the discharged steam is greater than an energy input to the pressure vessel. The enthalpy output corresponding to the discharged steam may be the total enthalpy of steam discharged from the pressure vessel during the respective period, determined by reference to the respective storage pressure of the pressure vessel as the steam was discharged. It is equivalent to the change in enthalpy of the water within the pressure vessel owing to the discharge of the steam. The enthalpy output may also be referred to as an enthalpy loss corresponding to the discharged steam. The energy input to the pressure vessel is the total of the enthalpy of any subcooled water provided to the pressure vessel (i.e. newly provided during the depletion period), and energy input to water within the pressure vessel by heating with the heater.

[0011] The steam may be selectively discharged from the outlet by control of a control valve, for example a control valve disposed at or downstream of the outlet such that the steam discharged from the outlet passes through the control valve. In some examples as described herein, the control valve may be remote from the vessel, for example provided

on a condensate return line in a thermosiphon arrangement (in which case the steam does not pass through the control valve).

[0012] It may be that, during the depletion period, the total enthalpy output is greater than the total energy input, whereas an instantaneous enthalpy output at any moment may be less than an instantaneous energy input at the same moment.

[0013] The depletion period may otherwise be described as a period in which mass and energy (or enthalpy) variations of water within the pressure vessel are such that the storage pressure reduces by at least 1 MPa from the peak storage pressure. The mass and energy variations may be due to the discharge of steam, any energy input by heating and/or any supply of subcooled water.

[0014] It may be that the discharged steam is provided to the thermal load directly without passing through an intermediate steam accumulator; or any steam accumulator or steam accumulators between the pressure vessel and the thermal load have a total volume which is less than a volume of the pressure vessel.

[0015] Herein, the expression “directly” is to be interpreted in view of the purpose of not passing through an accumulator. Accordingly, the discharged steam may be provided to the thermal load via one or more other components (e.g. via a desuperheater), but may not pass through an accumulator.

[0016] The method may comprise operating the heater during the depletion period. The heater may be operated during the depletion period simultaneously with selectively discharging steam.

[0017] The method may comprise operating the heater without simultaneously providing subcooled water to the pressure vessel; for part or all of the depletion period; and/or for part or all of a recharge period, in which the storage pressure increases by at least 1 MPa to a peak storage pressure of at least 2 MPa.

[0018] The recharge period as referred to herein may be a period in which an energy input to pressure vessel by heating is greater than an enthalpy output corresponding to steam discharge to meet the thermal energy demand. The recharge period may otherwise be described as a period in which the mass and energy (or enthalpy) variations of water within the pressure vessel are such that the storage pressure rises by at least 1 MPa to the peak storage pressure. The mass and energy variations may be due to heating, any discharge of steam and/or any supply of subcooled water.

[0019] It may be that steam is discharged from the outlet so that a liquid level of liquid water (i.e. a surface of the liquid water, or the interface between liquid water and steam) within the pressure vessel falls below a lower limit liquid level for operation of the heater during the depletion period. It may be that the controller operates the heater during the depletion period simultaneously with steam being discharged when the liquid level of water is above the lower limit liquid level, and that the controller deactivates the heater for continued discharge of steam beyond the lower limit liquid level.

[0020] The liquid level of liquid water within the pressure vessel may fall below the lower limit liquid level for operation of the heater during an extended portion of the depletion period. “Extended” signifies that the ability to continue discharging steam is extended by permitting the liquid level to fall below the lower limit liquid level, with the

trade-off that the heater cannot be operated until the liquid level is raised once again by providing additional water into the pressure vessel. After the extended depletion period is over (e.g. once discharge of steam is stopped, for example upon reaching the lower limit pressure condition), the pressure vessel may be provided with water to restore the liquid level to above the liquid level limit, and then the heater may be activated once again.

[0021] For example, the lower limit liquid level may be a lower limit liquid level height for operation of the heater, (i.e. a minimum height of the interface between the liquid water and steam), which may correspond to or be the height to which any heating element of the heater extends within the vessel. The lower limit liquid level may be a lower limit liquid volume of liquid water, which may be expressed as a volume fraction of liquid water within the pressure vessel (i.e. from 0 to 1).

[0022] In other words, steam may be discharged so that the liquid level margin becomes negative.

[0023] It may be that the steam is discharged to the thermal load at a discharge pressure; and subcooled water may be provided to the vessel when the storage pressure is greater than the discharge pressure.

[0024] The subcooled water is subcooled at the moment of supply to the pressure vessel (e.g. from a condensate supply vessel). In other words, the water is supplied from a condensate supply vessel which stores the condensate at a lower temperature than a temperature of water in the pressure vessel.

[0025] The discharge pressure may be variable and subcooled water may be provided when steam is not being discharged, in which case the discharge pressure is to be understood as the discharge pressure to which steam was last discharged.

[0026] It may be that the peak storage pressure exceeds a maximum water supply pressure at which subcooled water is provided to the vessel.

[0027] It may be that subcooled water is provided to the pressure vessel during the depletion period. It may be that a mass of subcooled water provided to the pressure vessel during the depletion period is no more than 50% of a mass of steam discharged during the depletion period, for example no more than 25%, no more than 10% or no more than 5%.

[0028] It may be that the thermal load discharges a flow of condensate corresponding to the discharged steam to a subcooled water supply vessel, such as a hotwell. It may be that a mass of condensate corresponding to at least 50% of a mass of steam discharged during the depletion period is provided to the subcooled water supply during the depletion period for subsequent re-supply to the pressure vessel; for example at least 75%, at least 90%, or substantially all of the condensate resulting from steam discharged from the pressure vessel during the depletion period.

[0029] The thermal load may comprise a heat exchanger oriented with respect to the pressure vessel so that a thermosiphon is established between the pressure vessel and the heat exchanger, such that discharged steam condensing within the heat exchanger forms a column of subcooled water that is returned to a condensate inlet at a lower portion of the pressure vessel, during the depletion period.

[0030] It may be that the heat exchanger is disposed at a higher position than the condensate inlet to give adequate head to return the subcooled water to the pressure vessel under the action of gravity. It may be that the heat exchanger

subcools the water within the heat exchanger by at least 10° C., relative to a saturation temperature corresponding to the storage pressure.

[0031] It may be that a control valve for controlling the discharge of steam from the pressure vessel is a condensate line control valve provided on a condensate line that returns condensed water from the heat exchanger to the pressure vessel, the control valve being controllable by the controller to vary or stop a flow rate of condensate from the heat exchanger to the pressure vessel, thereby also controlling steam discharge from the pressure vessel.

[0032] The subcooled water may be provided to the pressure vessel at an inlet flow rate such that the storage pressure reduces as the subcooled water is provided.

[0033] A flash potential corresponds to an amount of steam that can be flashed from liquid water within the pressure vessel before the storage pressure reaches a lower limit pressure for sustaining discharge of steam to the thermal load.

[0034] A liquid level margin corresponds to an amount of liquid water within the pressure vessel above a lower limit amount for operation of the heater upon discharge of the flash potential.

[0035] It may be that the method comprises evaluating a criterion corresponding to whether the liquid level margin is positive or negative; and based on the evaluation, providing subcooled water to the pressure vessel to increase the liquid level margin and to reduce the flash potential.

[0036] The subcooled water may be provided to increase the liquid level margin and reduce the flash potential when the evaluation corresponds to the liquid level margin being negative.

[0037] By way of example, the flash potential may correspond to or be defined as a mass of steam that can be flashed from liquid water within the pressure vessel before the storage pressure reaches a lower limit pressure for sustaining discharge of steam to the thermal load. The flash potential may be defined in other ways, such as by reference to a time for which a predetermined peak discharge rate of steam to the thermal load can be maintained.

[0038] By way of example, the liquid level margin may correspond to or be defined as a difference between a height of a liquid level of water within the pressure vessel upon discharge of the flash potential and a lower limit liquid level height for operation of the heater (the difference being negative when the liquid level is below the lower limit liquid level). The liquid level margin may be defined as a difference between a volume of liquid water within the pressure vessel upon discharge of the flash potential and a lower limit liquid volume of liquid water for operation of the heater. The volume may be expressed as a volume fraction of liquid water within the pressure vessel (i.e. from 0 to 1).

[0039] The criterion may correspond to whether the liquid level margin is positive or negative based on discharge of an instantaneous flash potential or based on discharge of a predicted flash potential.

[0040] Evaluating the criterion may comprise determining the flash potential, and determining the liquid level margin based on discharge of the flash potential (e.g. an instantaneous flash potential or a predicted flash potential). An instantaneous flash potential may be a function of a current storage pressure and liquid level of water within the pressure vessel, and the lower limit pressure.

[0041] The flash potential may be a predicted flash potential which is a function of a predicted demand and/or a predicted power output of the heater over a time period. The liquid level margin may be a predicted liquid level margin based on discharge of the predicted flash potential over the time period.

[0042] For example, evaluating the criterion may comprise evaluating a criterion corresponding to whether the predicted liquid level margin is positive or negative based on discharge of the predicted flash potential over the time period. Evaluating the criterion may comprise determining the predicted flash potential, determining the predicted liquid level margin based on discharge of the predicted flash potential over the time period, and determining whether the predicted liquid level margin is positive or negative.

[0043] The predicted flash potential may be determined based on a current liquid level and storage pressure, and based on the predicted demand and/or the predicted power output of the heater over the time period. The demand may be a predicted thermal energy demand or a predicted steam demand, for example.

[0044] The predicted demand may be based on historical demand data, such as historical steam or thermal demand data for an installation; and/or the predicted demand may be based on forecast weather conditions; and/or the predicted power output of the heater may be based on historical power availability data for the heater; and/or the predicted power output of the heater may be based on forecast weather conditions.

[0045] There may be a plurality of thermal loads including a first thermal load and a second thermal load. Steam may be selectively discharged from the pressure vessel to each of the thermal loads via respective control valves, based on respective thermal energy demands. The method may comprise evaluating a criterion corresponding to whether the flash potential is sufficient to meet a predicted demand of the plurality of loads. The method may comprise, based on the evaluation and on priority data relating to the thermal loads, determining to discharge steam to the first thermal load to meet a respective first thermal energy demand in preference to discharging steam to the second thermal load to meet a respective second thermal energy demand.

[0046] The method may comprise, subsequent to the determination:

[0047] discharging steam to the first thermal load to meet the respective first thermal energy demand; and

[0048] discharging steam to the second thermal load to only partially meet the respective second thermal energy demand; or

[0049] preventing discharge of steam to the second thermal load irrespective of the respective second thermal energy demand; or

[0050] communicating a load shedding signal to a controller that controls at least the second thermal load to indicate a reduced capacity to meet the second thermal energy demand; whereby the controller may operate the second thermal load to reduce the respective thermal energy demand.

[0051] The method may comprise, during a recharge period: heating liquid in the pressure vessel to raise the storage pressure by at least 1 MPa to a peak storage pressure of at least 2 MPa; providing subcooled water to the pressure vessel to reach a peak mass of water in the pressure vessel corresponding to a peak liquid level at the peak storage

pressure. A duration of the recharge period may be at least as 100% of the duration of the depletion period, for example at least 125% or at least 150%.

[0052] For example, the peak liquid level at the peak storage pressure may be 90%, or may be at least 80%.

[0053] The method may comprise phasing a profile of water supply during the recharge period so that it is front-loaded relative to a profile of heating during the recharge period.

[0054] By front-loading the supply of subcooled water, a temperature of water within the pressure vessel can be maintained relatively low during the recharge period (for example, as compared with a uniform profile of water supply or a back-loaded profile of water supply), thereby reducing thermal losses through a wall of the pressure vessel.

[0055] Phasing the supply of water and heating so that the profile of water is front-loaded relative to the profile of heating may comprise maintaining a minimum recharge flash potential (which may be a predetermined minimum flash potential or a flash potential corresponding to a predicted demand), while a liquid level margin progressively increases to the peak mass of water; and subsequently heating the liquid water to raise the pressure to the peak storage pressure.

[0056] The flash potential may be as defined elsewhere herein. The liquid level margin may be as defined elsewhere herein.

[0057] A flash potential corresponds to an amount of steam that can be flashed from liquid water within the pressure vessel before the storage pressure reaches a lower limit pressure for sustaining discharge of steam to the thermal load.

[0058] A liquid level margin corresponds to an amount of liquid water within the pressure vessel above a lower limit amount for operation of the heater upon discharge of the flash potential.

[0059] The method may comprise, during a water-priority portion of the recharge period, phasing a profile of water supply relative to a profile of heating to maintain a minimum recharge flash potential while increasing an amount of water within the pressure vessel to the target peak mass of water, such that the liquid level margin progressively rises; and subsequently heating the liquid water during a flash-priority portion of the recharge period to raise the storage pressure to the peak storage pressure.

[0060] The minimum recharge flash potential may be a predetermined minimum flash potential or a flash potential corresponding to a predicted demand.

[0061] The minimum recharge flash potential may correspond to a predicted demand. The predicted demand may correspond to (e.g. be) no demand for steam during an idle portion of the recharge period, and the method may comprise providing subcooled water to reduce the storage pressure below the lower limit pressure during the idle portion, the pressure subsequently rising owing to heat input to provide a flash potential corresponding to a non-zero demand.

[0062] A maximum flow rate of steam discharge during the depletion period per unit area of a liquid level of water in the pressure vessel is no more than 150 kg/m²h, for example no more than 100 kg/m²h or no more than 50 kg/m²h. The maximum flow rate of steam discharge may per unit area may be expressed as a percentage of the maximum steam release rate without steam entrainment (the MSR) as

explained elsewhere herein, and may be no more than 10% of the MSR, for example no more than 5% or no more than 3%.

[0063] The method may comprise deaerating an inlet flow of subcooled water provided to the pressure vessel along a deaeration path by conveying a deaerating flow of steam from within the pressure vessel past in counterflow along the deaeration path. The inlet flow may be provided into the pressure vessel over a range of storage pressures of the pressure vessel. A velocity of the deaerating flow of steam is varied as a function of a temperature of the inlet flow and/or a temperature difference between the inlet flow and the steam to target the inlet flow reaching a saturation temperature corresponding to the steam over the range of storage pressures. The velocity may be varied by controlling a control valve for venting the deaerating flow of steam and associated entrained gases. The velocity or a control valve setting for controlling the velocity may be determined by reference to a database of velocities or control valve settings correlated to the temperature of the inlet flow and/or the temperature of the steam.

[0064] An inlet flow of subcooled water may be deaerated during a depletion period or a recharge period as described herein.

[0065] It may be that the peak storage pressure is at least 2.5 MPa, for example at least 3 MPa; and/or it may be that the storage pressure reduces during the depletion period to a value of no more than 1.5 MPa, for example no more than 1 MPa, or no more than 0.8 MPa.

[0066] An average depletion power may be defined as a cumulative enthalpy of steam discharged from the pressure vessel during the depletion period divided by a duration of the depletion period. A maximum depletion power may be defined as a maximum enthalpy of steam discharged within the depletion period during any minimum power evaluation period of one minute, divided by the minimum power evaluation period.

[0067] The method may further comprise: heating liquid water within the pressure vessel, using the heater, during a recharge period in which the storage pressure increases by at least 1 MPa. An average recharge power may be defined as a cumulative energy provided to the liquid water by the heater during the recharge period, divided by a duration of the recharge period. A maximum reheat power may be defined as a maximum power at which energy is provided to the liquid water by the heater within the recharge period. It may be that the average reheat power is no more than 50% of the average depletion power; and/or the average reheat power is no more than 50% of the maximum depletion power; and/or the maximum reheat power is no more than 50% of the maximum depletion power.

[0068] It may be that a dimensional ratio of the cumulative enthalpy of (i) steam discharged from the pressure vessel during the depletion period and (ii) the average reheat power of the heater during the recharge period is at least 25000 seconds.

[0069] The method may be conducted using a thermal installation in accordance with the second aspect of the disclosure.

[0070] According to a second aspect there is provided a thermal energy storage and supply installation comprising:

[0071] a pressure vessel for storing water comprising saturated liquid water and steam at a storage pressure of 2 MPa, the pressure vessel having an outlet to discharge steam to a thermal load;

[0072] an electrically-powered heater configured to heat liquid water stored in the pressure vessel to vary a storage pressure within the pressure vessel;

[0073] a controller configured to operate the thermal storage installation by:

[0074] controlling the heater to heat liquid water within the pressure vessel to reach a peak storage pressure of saturated liquid water and steam of at least 2 MPa;

[0075] controlling a control valve to selectively discharge steam from the outlet to a thermal load, in response to a thermal energy demand;

[0076] permitting steam discharge to meet the thermal energy demand so that the storage pressure reduces by at least 1 MPa from the peak storage pressure.

[0077] The peak storage pressure may be at least 2.5 MPa, for example at least 3 MPa.

[0078] The steam may be selectively discharged from the outlet by control of a control valve, for example the steam may be discharged to the thermal load via a control valve. The control valve may be disposed at or downstream of the outlet such that the steam discharged from the outlet passes through the control valve. In some examples as described herein, the control valve may be remote from the vessel, for example provided on a condensate return line in a thermosiphon arrangement (in which case the steam does not pass through the control valve).

[0079] It may be that the outlet is in communication with the thermal load to provide the discharged steam directly to the thermal load without passing through an intermediate steam accumulator; or any steam accumulator or steam accumulators between the pressure vessel and the thermal load may have a total volume which is less than a volume of the pressure vessel.

[0080] It may be that the controller is configured to control the heater to heat liquid water within the pressure vessel independently of controlling the control valve to selectively discharge steam, whereby in use simultaneous heating and discharge of steam is permitted, and each of heating and discharge of steam is permitted without the other.

[0081] It may be that the controller is configured to control the heater to heat liquid water within the pressure vessel independently of causing subcooled water to be supplied to the pressure vessel, whereby in use simultaneous heating and subcooled water supply is permitted, and heating without simultaneous subcooled water supply is permitted.

[0082] It may be that the controller is configured to cause subcooled water to be supplied to the pressure vessel when the storage pressure is greater than a discharge pressure to which steam is discharged from the outlet (e.g. via a control valve).

[0083] The installation may further comprise a water pump configured to supply subcooled water to the pressure vessel; optionally wherein the water pump has a maximum water supply pressure which is lower than the peak storage pressure.

[0084] The installation may further comprise a subcooled water supply vessel configured to store subcooled water for supply to the pressure vessel. A ratio of a storage volume of

the subcooled water supply vessel to a storage volume of the pressure vessel may be at least 5%; optionally at least 7.5% or at least 10%.

[0085] It may be that the controller is configured to selectively operate in an extended depletion mode in which the controller permits steam to be discharged so that a liquid level of liquid water within the pressure vessel falls below a lower limit liquid level for operation of the heater during the depletion period, wherein the controller prevents heating with the heater when the liquid level is below the lower limit liquid level for operation of the heater. It may be that the controller is configured to selectively operate the heater in a heated depletion mode in which the controller prevents steam discharge that would cause the liquid level to fall below the lower limit liquid level; wherein the controller permits heating with the heater when the liquid level is below the lower limit liquid level. It may be that the thermal load comprises a heat exchanger oriented with respect to the pressure vessel to define a thermosiphon between the pressure vessel and the heat exchanger, whereby discharged steam condensing within the heat exchanger forms a column of subcooled water that is returned to a condensate inlet at a lower portion of the pressure vessel. The thermal load may be part of the installation.

[0086] It may be that the heat exchanger is disposed at a higher position than the pressure vessel and the controller is configured to cause steam to be discharged to the heat exchanger at a discharge pressure selected such that, given the relative position of the heat exchanger, there is adequate head to return the subcooled water to the pressure vessel under the action of gravity. The controller may be configured to control heat exchange at the heat exchanger such that the water is subcooled by at least 10° C.

[0087] The installation may comprise a level sensor configured to provide a level signal corresponding to a liquid level of water within the pressure vessel to the controller; and/or a storage pressure sensor configured to provide a pressure signal corresponding to a pressure of water within the pressure vessel to the controller, and/or a storage temperature signal configured to provide a temperature signal corresponding to a temperature of water within the pressure vessel to the controller; and/or a discharge pressure sensor configured to provide a pressure signal to the controller corresponding to a pressure to which steam discharged from the outlet of the pressure vessel (e.g. through a respective control valve); and/or a discharge flowmeter configured to provide a flow rate signal to the controller corresponding to a flow rate of steam downstream of the outlet (e.g. downstream of a respective control valve); and/or an inlet flowmeter configured to provide a flow rate signal to the controller corresponding to a flow rate of subcooled water provided to the pressure vessel.

[0088] A flash potential may correspond to an amount of steam that can be flashed from liquid water within the pressure vessel before the storage pressure reaches a lower limit pressure for sustaining discharge of steam to the thermal load.

[0089] A liquid level margin may correspond to an amount of liquid water within the pressure vessel above a lower limit amount for operation of the heater upon discharge of the flash potential.

[0090] It may be that the controller is configured to select between operating the thermal storage installation in a recharge mode or a depletion mode based on a predeter-

mined setting such as a time-dependent setting, based on a user input, or based on a predicted demand profile and/or a predicted power output profile of the heater. It may be that, in the depletion mode, the controller is configured to evaluate a criterion corresponding to whether the liquid level margin is positive or negative; and to provide subcooled water to the pressure vessel to increase the liquid level margin and reduce the flash potential when the evaluation corresponds to the liquid level margin being negative. It may be that, in the recharge mode, the controller is configured to phase a profile of water supply relative to a profile of heating to (i) maintain a minimum recharge flash potential which is a predetermined flash potential or which corresponds to a predicted demand, while a liquid level margin and an amount of water in the pressure vessel progressively increases to a target mass of water corresponding to a target liquid level at the peak storage pressure; and (ii) subsequently heat the liquid water to raise the storage pressure to the peak storage pressure.

[0091] A useful flash potential corresponds to an amount of steam that can be flashed from liquid water within the pressure vessel before any of (i) the storage pressure reaches the lower limit pressure for sustaining discharge of steam to the thermal load and (ii) an amount of liquid water within the pressure vessel reaches the lower limit amount. The supply of subcooled water in the depletion mode to increase the liquid level margin and reduce the flash potential when the evaluation corresponds to the liquid level margin being negative may be to increase the useful flash potential.

[0092] The criterion may correspond to whether the liquid level margin is positive or negative based on discharge of an instantaneous flash potential or based on discharge of a predicted flash potential.

[0093] The controller may be configured to evaluate the criterion in the depletion mode, and/or phase the profile of water supply in the recharge mode, by determining the flash potential, and determining the liquid level margin based on discharge of the flash potential (e.g. an instantaneous flash potential or a predicted flash potential). An instantaneous flash potential may be a function of a current storage pressure and liquid level of water within the pressure vessel, and the lower limit pressure. Otherwise, the controller may be configured to evaluate the criterion and/or phase the profile of water supply in the recharge without directly determining the flash potential and liquid level margin, for example based on a level signal received from a level sensor as described elsewhere herein.

[0094] It may be that the flash potential is a predicted flash potential, and wherein the controller is configured to predict the flash potential as a function of a predicted demand and/or a predicted power output of the heater over a time period. It may be that the liquid level margin is a predicted liquid level margin, wherein the controller is configured to predict the liquid level margin based on discharge of the predicted flash potential over the time period.

[0095] The evaluated criterion may be a criterion corresponding to whether the predicted liquid level margin is positive or negative based on discharge of the predicted flash potential over the time period. The controller may be configured to evaluate the criterion by determining the predicted flash potential, determining the predicted liquid level margin based on discharge of the predicted flash potential over the time period, and determining whether the predicted liquid level margin is positive or negative.

[0096] The controller may be configured to determine the predicted flash potential based on a current liquid level and storage pressure, and based on the predicted demand and/or the predicted power output of the heater over the time period. The demand may be a predicted thermal energy demand or a predicted steam demand, for example.

[0097] The predicted power output profile of the heater may correspond to a predicted power availability for heating using the heater.

[0098] It may be that the controller is configured to determine the predicted demand based on historical demand data, such as historical steam or thermal demand data for the installation; and/or the controller is configured to determine the predicted demand based on forecast weather conditions; and/or the controller is configured to determine the predicted power output of the heater based on historical power availability data for the heater; and/or the controller is configured to determine the predicted power output of the heater based on forecast weather conditions.

[0099] It may be that, in the recharge mode, the controller is configured so that when the predicted demand corresponds to no demand for steam during an idle portion of a recharge period, the controller causes subcooled water to be supplied to reduce the storage pressure below the lower limit pressure during the idle portion. It may be that, the controller is configured to subsequently raise the storage pressure by heating liquid water within the pressure vessel to provide a flash potential corresponding to a non-zero demand.

[0100] It may be that the controller is configured to control the control valve so that steam is selectively discharged at a maximum flow rate per unit area of a liquid level (i.e. surface) of water in the pressure vessel which is no more than $150 \text{ kg/m}^2\text{h}$, for example no more than $100 \text{ kg/m}^2\text{h}$ or no more than $50 \text{ kg/m}^2\text{h}$. The maximum flow rate of steam discharge may per unit area may be expressed as a percentage of the maximum steam release rate without steam entrainment (the MSR) as explained elsewhere herein, and may be no more than 10% of the MSR, for example no more than 5% or no more than 3%.

[0101] It may be that the pressure vessel is configured to discharge steam to a plurality of thermal loads via respective control valves, including a first thermal load and a second thermal load. The controller may be configured to evaluate a criterion corresponding to whether the flash potential is sufficient to meet a predicted demand of the plurality of loads. The controller may be configured so that responsive to an outcome of the evaluation, the controller evaluates priority data specifying relative priorities of the thermal loads, and operates in a priority discharge mode in which it controls selective discharge of steam to meet a thermal energy demand of a relatively high priority thermal load in preference to controlling selective discharge of steam to meet a thermal energy demand of a relatively lower priority thermal load.

[0102] The outcome of the evaluation may be an outcome indicative of the flash potential being insufficient to meet the predicted demand.

[0103] In the priority discharge mode, the controller may be configured to control a control valve associated with the relatively higher priority load to discharge steam to meet a respective thermal energy demand (e.g. as determined based on a respective demand signal received by the controller), and to either: (i) control a control valve associated with the relatively lower priority load to discharge steam to only

partially meet a respective thermal energy demand (e.g. as determined based on a respective demand signal received by the controller); or (ii) control the control valve associated with the relatively lower priority load to prevent discharge of steam to the load; and/or (iii) transmit a load-shedding signal to a controller of the relatively lower priority thermal load to indicate a reduced capacity to meet the respective thermal energy demand.

[0104] It may be that the installation is configured to provide water in the form of steam discharged from the pressure vessel to the thermal load in an open arrangement, by which the thermal load consumes the water provided as steam without corresponding return of the water as condensate. The controller may be configured to discharge steam to the thermal load at a minimum discharge pressure of between 0.5 MPa and 1.0 MPa, or the thermal load is configured to receive discharged steam at a minimum discharge pressure of between 0.5 MPa and 1.0 MPa.

[0105] It may be that the installation is configured to supply steam discharged from the pressure vessel to at least two thermal loads. The installation may be configured to provide water in the form of steam discharged from the pressure vessel to one of the thermal loads in an open arrangement, by which the water is placed into contact with a foreign process fluid or article, and/or is discharged from the thermal load without corresponding return of the water as condensate. Additionally or alternatively, it may be that the installation is configured to provide water in the form of steam discharged from the pressure vessel to one of the thermal loads in a closed loop, by which the water is at least partly returned to the pressure vessel as subcooled water, for example via a subcooled supply vessel.

[0106] In the closed loop, the water (provided as steam) may be at least partly returned to the pressure vessel as subcooled water without contact with a foreign process fluid or article.

[0107] It may be that the pressure vessel is provided with a deaerator configured to receive an inlet flow of subcooled water along a deaeration path, and configured to direct a deaerating flow of steam from within the pressure vessel in counterflow along the deaeration path. The controller may be configured to vary a velocity of the deaerating flow of steam as a function of a temperature of the inlet flow and/or a temperature of the steam, to target the inlet flow reaching a saturation temperature corresponding to the steam along the deaeration path, over a range of storage pressures. The controller may be configured to vary the velocity by controlling a control valve for venting the deaerating flow of steam and associated entrained gases from the deaerator. The controller may be configured to control the velocity or a control valve setting for controlling the velocity by reference to a database of velocities or control valve settings correlated to the temperature of the inlet flow and/or the temperature of the steam.

[0108] It may be that the heater is installed within the pressure vessel so that a lower limit liquid level for operation of the heater corresponds to a liquid fraction of water within the pressure vessel of no more than 60%, for example no more than 50%, no more than 40% or no more than 30%.

[0109] According to a third aspect there is disclosed a method of thermal storage and supply corresponding to the first aspect and differing from the first aspect with respect to the peak storage pressure and the reduction of pressure

during the depletion period (and other related numerical definitions as are described below). Statements relating to how the definition of the third aspect may differ from the first aspect are defined below. In all other respects, except where mutually exclusive, a feature described in relation to the first aspect may be applied mutatis mutandis to the third aspect.

[0110] According to the third aspect, the heater is controlled to raise the storage pressure to a peak storage pressure of at least 0.5 MPa, for example at least 1 MPa or at least 2 MPa. The selective discharge of steam is such that during the depletion period the storage pressure reduces by a depletion pressure difference which is at least 50% of the peak storage pressure.

[0111] In accordance with the third aspect, the depletion period may be described as a period in which mass and energy (or enthalpy) variations of water within the pressure vessel are such that the storage pressure reduces by the depletion pressure difference.

[0112] A definition with respect to a recharge period in the first aspect relates to an increase of the storage pressure. A corresponding definition with respect to the third aspect corresponds to a recharge pressure difference which is at least 50% of the peak storage pressure. In particular, the method may comprise operating the heater without simultaneously providing subcooled water to the pressure vessel; for part or all of the depletion period; and/or for part or all of a recharge period, in which the storage pressure increases by a recharge pressure difference to the peak storage pressure, wherein the recharge pressure difference is at least 50% of the peak storage pressure. As stated above, the peak storage pressure may be at least 0.5 MPa, for example at least 1 MPa or at least 2 MPa.

[0113] In accordance with the third aspect, the recharge period may be described as a period in which the mass and energy (or enthalpy) variations of water within the pressure vessel are such that the storage pressure rises by the recharge pressure difference to the peak storage pressure.

[0114] In accordance with the third aspect, the method may comprise, during a recharge period: heating liquid in the pressure vessel to raise the storage pressure by the recharge pressure difference to the peak storage pressure; providing subcooled water to the pressure vessel to reach a peak mass of water in the pressure vessel corresponding to a peak liquid level at the peak storage pressure. A duration of the recharge period may be at least 100% of the duration of the depletion period, for example at least 125% or at least 150%.

[0115] A method according to the third aspect may be conducted using a thermal installation in accordance with the second aspect or a thermal energy storage and supply apparatus according to the fourth aspect.

[0116] According to a fourth aspect there is provided a thermal energy storage and supply apparatus (e.g. an installation) corresponding to the installation of the second aspect and differing from the second aspect with respect to the peak storage pressure and the reduction of pressure during the depletion period (and other related numerical definitions as are defined below). Statements relating to how the definition of the fourth aspect may differ from the second aspect are defined below. In all other respects, except where mutually exclusive, any feature described in relation to any of the first, second and third aspects may be applied mutatis mutandis to the fourth aspect.

[0117] According to the fourth aspect, the thermal energy storage and supply apparatus comprises:

[0118] a pressure vessel for storing water comprising saturated liquid water and steam at a storage pressure of at least 0.5 MPa, for example at least 1 MPa or at least 2 MPa, the pressure vessel having an outlet to discharge steam to a thermal load;

[0119] an electrically-powered heater configured to heat liquid water stored in the pressure vessel to vary a storage pressure within the pressure vessel;

[0120] a controller configured to operate the thermal storage apparatus by:

[0121] controlling the heater to heat liquid water within the pressure vessel to reach a peak storage pressure of saturated liquid water and steam of at least 0.5 MPa, for example at least 1 MPa or at least 2 MPa;

[0122] controlling a control valve to selectively discharge steam from the outlet to a thermal load, in response to a thermal energy demand;

[0123] permitting steam discharge to meet the thermal energy demand so that the storage pressure reduces from the peak storage pressure by at least 50% of the peak storage pressure.

[0124] A definition with respect to a discharge pressure is provided with respect to the second aspect. It may be that a thermal energy storage and supply apparatus according to the fourth aspect is configured to discharge steam at a variable discharge pressure (which may be the same or lower than a prevailing storage pressure). The controller(s) described herein may comprise a processor. The controller and/or the processor may comprise any suitable circuitry to cause performance of the methods described herein and as illustrated in the drawings. The controller or processor may comprise: at least one application specific integrated circuit (ASIC); and/or at least one field programmable gate array (FPGA); and/or single or multi-processor architectures; and/or sequential (Von Neumann)/parallel architectures; and/or at least one programmable logic controllers (PLCs); and/or at least one microprocessor; and/or at least one microcontroller; and/or a central processing unit (CPU), to perform the methods and or stated functions for which the controller or processor is configured.

[0125] The controller may comprise or the processor may comprise or be in communication with one or more memories that store that data described herein, and/or that store machine readable instructions (e.g. software) for performing the processes and functions described herein (e.g. determinations of parameters and execution of control routines).

[0126] The memory may be any suitable non-transitory computer readable storage medium, data storage device or devices, and may comprise a hard disk and/or solid state memory (such as flash memory). In some examples, the computer readable instructions may be transferred to the memory via a wireless signal or via a wired signal. The memory may be permanent non-removable memory, or may be removable memory (such as a universal serial bus (USB) flash drive). The memory may store a computer program comprising computer readable instructions that, when read by a processor or controller, causes performance of the methods described herein, and/or as illustrated in the Figures. The computer program may be software or firmware, or be a combination of software and firmware.

[0127] Except where mutually exclusive, a feature described in relation to any one of the above aspects may be applied mutatis mutandis to any other aspect. Furthermore, except where mutually exclusive, any feature described herein may be applied to any aspect and/or combined with any other feature described herein.

BRIEF DESCRIPTION OF THE DRAWINGS

[0128] The invention will be described with reference to the accompanying drawings, in which:

[0129] FIG. 1 schematically shows an example thermal energy storage and supply installation for an industrial plant;

[0130] FIG. 2 shows an example profile of steam demand;

[0131] FIGS. 3a, 3b, 4a, 4b show example profiles of operating conditions of a thermal energy storage and supply installation;

[0132] FIG. 5 is a flow diagram of a method of controlling steam discharge;

[0133] FIG. 6 is a flow diagram of a method of controlling heating;

[0134] FIGS. 7a-7e schematically show various liquid levels within a pressure vessel with reference to illustrate the concept of a flash potential;

[0135] FIG. 8 is a flow diagram of a method of controlling supplying subcooled water to a pressure vessel; and

[0136] FIG. 9 schematically shows a further example of a thermal energy storage and supply installation.

DETAILED DESCRIPTION

[0137] FIG. 1 schematically shows an industrial plant 10 as an example of a facility having at least one thermal load, provided with a thermal energy storage and supply installation 100.

[0138] The industrial plant 10 has first and second plant thermal loads 30, 40 which in this particular example are a space heating system and a sterilization system respectively, but in variant examples may be any system that demands thermal energy. The first plant thermal load 30 (space heating system) is configured to receive thermal energy (heat) via a heat transfer circuit 132 extending between the installation 100 and the industrial plant 10, rather than directly receiving steam from the installation 100 as a process fluid for the thermal load, as will be further described below. In contrast, the second plant thermal load 40 (the sterilization system) is configured to directly receive steam discharged from the installation 100 for use as a process fluid of the thermal load itself, e.g. for sterilization of articles. Steam consumed in the second plant thermal load 40 (i.e. condensed to give up its latent heat) is discharged to a drain 50, but in variant examples may be returned as condensate to a water supply of the installation 100.

[0139] In this example, the plant thermal loads 30, 40 are operated independently of the installation 100, and are operatively coupled with a controller 200 of the installation (as will be described below) to communicate respective thermal energy demands. The thermal energy demand may be communicated in any suitable form, for example as a demand signal corresponding to a thermal power requirement (e.g. in kW), a steam demand (e.g. in kg/s), or a related parameter such as a percentage of a predetermined maximum demand of any type (e.g. encoding “0.5” to be interpreted by the controller as a proportion of a maximum demand such as 300 kW, to result in a demand of 150 kW).

[0140] The thermal storage and supply system 100 comprises a pressure vessel 110 for storing water at high pressure and temperature. For completeness, the word “water” is used in this disclosure to refer to water in any phase, whereas the expressions “liquid water” and “steam” are used to refer to water in the liquid and gas phases respectively. The pressure vessel 110 is configured to store saturated liquid water and steam at a storage pressure of at least 2 MPa, for example at least 3 MPa, or at least 4 MPa, or between 2 MPa to 6 MPa. In the particular examples described below the pressure vessel 110 is used to store water up to a target peak storage pressure of approximately 3 MPa (e.g. 3.101 MPa corresponding to 30 bar gauge pressure for an installation situated at atmospheric pressure). It will be appreciated that these values correspond to a minimum requirement of the pressure vessel, and the pressure vessel can of course store water at lower values of storage pressure. As is discussed elsewhere herein, in variant examples the pressure vessel may be configured to store saturated liquid water and steam at a lower storage pressure (serving as a correspondingly lower minimum requirement as to the storage capability), for example at least 0.5 MPa or at least 1 MPa.

[0141] The pressure vessel 110 comprises an electrically-powered heater 112 configured to heat liquid water stored in the pressure vessel to vary a storage pressure within the pressure vessel. The heater 112 is coupled to an electrical power source 113, which may comprise a local power source such as a renewable energy source (e.g. a solar electric power source or a wind electric power source), and/or a generator (e.g. coupled to an engine such as a gas turbine). The electrical power source may comprise a local power source and/or may comprise an electrical grid connection for providing power from a non-local electrical grid (e.g. a national grid). The local power source may be a microgrid. A microgrid may be defined as a local power distribution network of a plurality of energy sources (e.g. electrical energy sources), that may be operated independently of a non-local electrical grid to provide power to a dedicated local geographic area, and optionally may connect to a non-local electrical grid. A microgrid may have defined boundaries and act as a single controllable entity. In contrast, a non-local electrical grid will be understood to serve a large geographical area (such as a state) and comprise a large number of power sources such that the grid as a whole is not controlled as a single entity. When the electrical power source of the installation comprises (i) a local power source which is a microgrid, and (ii) a grid connection to a non-local electrical grid, the grid connection may be a connection of the microgrid to the non-local electrical grid. While a person skilled in the art understands the difference in scale and control of a microgrid and a non-local electrical grid, it can be stated that when the electrical power source comprises (i) a local power source comprising a microgrid and (ii) a grid connection to a non-local electrical grid, a power generation capacity of the microgrid (when operated independently of any non-local electrical grid) is less than a power generation capacity of the non-local electrical grid. For example, it may be no more than 10%, no more than 1% or no more than 0.1% of the power generation capacity of the non-local electrical grid.

[0142] The pressure vessel 110 may provide energy storage for the microgrid, in the form of thermal energy storage that may be distributed to local thermal loads.

[0143] In this example the heater **112** is located in a lower portion of the pressure vessel **110**, such that lower limit liquid level of water in the pressure vessel to submerge the heater (or at least each heating element of the heater) is a height of approximately 33% of the diameter of the pressure vessel, corresponding to a liquid fraction of approximately 30%. In other examples the lower limit liquid level may be higher or lower, for example it may correspond to liquid fraction of between 20% and 70%, for example 20%-60% or 25%-50%. A liquid level **114** of water within the pressure vessel is the interface between liquid water and gas, such as saturated liquid water and steam.

[0144] The pressure vessel **110** is provided with a sensor **210** coupled to the controller **200** and configured to communicate a storage pressure signal corresponding to a storage pressure of water within the pressure vessel **110** to the controller **200**. As will be appreciated, when water is saturated such that the vessel comprises liquid water and steam, the pressure directly corresponds to a respective saturation temperature. As such, a suitable sensor may be a pressure sensor configured to monitor pressure or a temperature sensor configured to monitor temperature. In this particular example, the sensor **210** is a pressure sensor configured to monitor the storage pressure and send a storage pressure signal encoding the monitored pressure to the controller **200**.

[0145] The pressure vessel **110** is further provided with a level sensor **214** which may be of any suitable type as known in the art, for example a capacitance probe level sensor having an elongate probe extend over a monitored range of the sensor and configured to output a continuously variable level signal (e.g. in mA) which, with suitable calibration at the level sensor **214** or the controller, can be processed to determine the liquid level of water in the pressure vessel throughout the monitored range. In other examples, a level sensor may comprise a plurality of sensors, for example each being configured to determine whether the liquid level is above or below a respective height/level within the pressure vessel, such that the level is determined as a discontinuous output quantity. The level sensor **214** is configured to communicate a level signal corresponding to the liquid level to the controller **200**.

[0146] The pressure vessel **110** comprises an outlet **116** for discharging steam from the pressure vessel via a control valve, which is operable to reduce the steam to a lower downstream pressure (a discharge pressure). In this example the outlet is located within an upper portion of the vessel above a maximum operating liquid level for the pressure vessel. The maximum operating liquid level may be a target peak liquid level (as will be described further below) or may be marginally higher to accommodate inadvertent operation above the target peak liquid level. For example, the target peak liquid level may correspond to a target peak liquid fraction of 90% of the vessel volume. In this example, there are first and second discharge control valves **216**, **217** associated with respective thermal loads as will be described in detail below, each discharge control valve being downstream of the outlet **116** and connected to it by a discharge line **101**. However, in other examples there may be a single or common discharge control valve either downstream of the outlet **116** (e.g. at the location **215** marked in dashed lines in FIG. 1) or integrated with the outlet **116** of the pressure vessel **110**. Each discharge control valve **216**, **217** is operatively coupled to the controller **200** to receive a respective discharge control signal to operate the valve.

[0147] The pressure vessel **110** is further provided with a deaerator **120**, which may be an integrated deaerator (i.e. integrated with the pressure vessel **110**) as shown in FIG. 1. The deaerator is configured to remove oxygen and other dissolved components such as carbon dioxide from a supply of liquid water provided to the pressure vessel. The deaerator **120** may be of any suitable type as is known in the art, but in this example is schematically shown having a configuration comprising a plurality of deaeration trays **121** vertically distributed through the deaerator. The deaerator **120** defines a deaeration path **122** for an inlet flow of water (in this example, a convoluted path past the trays **121**) and is configured to direct a deaerating flow of steam from within the pressure vessel along the deaeration path in counterflow with the inlet flow. By passing the deaerating flow along the deaeration path, the inlet flow of water is raised to the saturation temperature of the vessel and dissolved substances such as oxygen and carbon dioxide are removed from the inlet flow to be discharged from the deaerator along with the deaerating flow. The deaerating flow may be discharged through a deaeration outlet provided with a deaerator outlet control valve **124** as shown in FIG. 1. The deaerator outlet control valve **124** is operatively coupled to the controller **200** to receive a deaeration flow control signal to operate the valve.

[0148] The inlet flow to the deaerator is controlled to selectively supply water from a water source to the pressure vessel in response to a demand, as will be described in detail below. In this example, the inlet flow is selectively caused to flow into the deaerator by an inlet pump **122**. The inlet pump **122** is operatively coupled to the controller **200** to receive a water supply signal by which it is controlled to provide the inlet flow at a variable inlet flow rate and at a suitable pressure to pass into the pressure vessel **110** and deaerator **120**. In variant examples, the inlet flow may be selectively permitted to flow into the deaerator by an inlet control valve that controls the inlet flow rate based on a water supply signal received from the controller **200**, and which may be downstream of an inlet pump **122** which pressurizes the water as described above, or which may be coupled to a supply or pressurized water without an intervening pump.

[0149] In variant examples, an inlet flow of water may be directly provided from a water source to the pressure vessel **110** (i.e. without passing through a deaerator that deaerates the inlet flow as it enters the pressure vessel), with the inlet flow being controlled by the controller **200** via an inlet pump and/or an inlet control valve as described above. In installations comprising a deaerator, there may be a bypass line between the water source and the pressure vessel which bypasses the deaerator, for example a bypass line extending from the inlet pump or from a valve arrangement (e.g. a three-way valve) downstream of the inlet pump so that an inlet flow of water can be selectively provided to the pressure vessel via the deaerator or bypassing the deaerator. Such an arrangement may be advantageous when the respective supply of water is a water supply vessel to which water is returned from within the installation in a closed loop (as will be described below), since the deaerator may be used for an initial supply of water to the pressure vessel that may require deaeration, and may subsequently be bypassed when the water supply vessel is determined to (or considered to) contain water that has already been deaerated.

[0150] A steam storage and supply installation as envisaged by the present disclosure may have an open arrangement or closed loop for the supply of steam to the thermal load.

[0151] An open arrangement is considered to be one where water (provided as steam) provided to the respective thermal load is placed into contact with a foreign process fluid or article, and/or is discharged from the thermal load without corresponding return of the water as condensate. The expression “a foreign process fluid or article” is intended to mean a substance that may contaminate the water provided to the thermal load. For example, when the water (provided as steam) is in direct contact with another fluid in the thermal load or downstream of it, it may mix with that fluid. Similarly, when the steam is used to heat foreign articles, such as articles to be sterilized in a sterilizer, or baked in a steam oven, the water (which may be steam) may be contaminated by substances from those articles. An alternative definition of the open arrangement would be an arrangement in which the water is not contained within a closed loop (a closed loop being configured to recirculate water to the pressure vessel without intervening treatment).

[0152] A closed loop is considered to be an arrangement whereby the water provided to the respective thermal load is partly or wholly returned to the pressure vessel as subcooled water, optionally via a subcooled supply vessel (which may discontinuously supply the water to the pressure vessel). The closed loop substantially prevents contamination of the water, such that the water does not contact a foreign process fluid or article.

[0153] The example installation 100 of FIG. 1 is configured to provide steam to a first thermal load 130 which is a heat exchanger in a closed loop arrangement, via a distribution line 102 downstream of the first discharge control valve 216. Water provided (as steam) to the first thermal load 130 is returned to the pressure vessel 110 via a water supply vessel 150 as will be described below. The heat exchanger 130 is thermally coupled to a first plant thermal load 30 of the industrial plant 10 via a process fluid circuit 132, without the steam provided to the heat exchanger 130 coming into direct contact with a process fluid of the process fluid circuit. The controller 200 is configured to activate a heat transfer pump 133 associated with the heat transfer circuit 132 upon selectively discharging steam to the heat exchanger 130 for heat transfer. In other examples there may be no process circuit 132, and the distribution line 102 may extend directly into the first plant thermal load 30, before returning to the water supply vessel 150 (still without direct contact with a process fluid of the first plant thermal load 30). The controller 200 is configured to operate the first discharge control valve 216 based on a demand signal received from the first plant thermal load 30 to control a flow rate of steam through the heat exchanger 130. In this example, the flow rate is monitored by a first flowmeter 218 disposed along the distribution line 102 which provides a first flow rate signal to the controller 200, and the controller 200 may control the first discharge control valve 216 based on the monitored flow rate in order to meet the thermal energy demand, and/or based on other parameters such as pressure and temperature as may be monitored by additional sensors along the distribution line 102.

[0154] The water supply vessel 150 has a return inlet for receiving water returned from the first thermal load 130 (the heat exchanger 130), and a source inlet for receiving water

from an external water supply (such as treated mains water or a condensate return system of the industrial plant 10). The water supply vessel 150 has an outlet configured to provide water to the pressure vessel 110. In this example the outlet provides water to the pressure vessel 110 via the inlet pump 122 and the deaerator 120, but as mentioned above in other examples water may be provided along a bypass line to the pressure vessel 110.

[0155] The example installation 100 is further configured to provide steam to a second thermal load, which is the sterilizer system 40 (the second plant thermal load 40) as described above, in an open arrangement. The steam is provided via a distribution line 103 downstream of the second discharge control valve 217. The controller 200 is configured to operate the second discharge control valve 217 based on a demand signal received from the second plant thermal load 40 to control a flow rate of steam to the second plant thermal load 40. It will be appreciated that the sterilizer 40 is a representative example of a thermal load of the industrial plant and in practice there may be a plurality of thermal loads configured to be supplied with steam by a steam network of the industrial plant which routes steam to each individual load. For the purposes of this disclosure, such a network and plurality of loads may be considered to be equivalent to a single second thermal load 40 having an associated total steam demand.

[0156] In this example, the flow rate to the second thermal load is monitored by a second flowmeter 219 disposed along the distribution line 103 which provides a second flow rate signal to the controller 200, and the controller 200 may control the second discharge control valve 217 based on the monitored flow rate in order to meet the thermal energy demand, and/or based on other parameters such as pressure and temperature as may be monitored by additional sensors along the distribution line 103. In this example, the second plant thermal load 40 is a sterilizer such that the steam is provided in an open arrangement by virtue of being in contact with foreign articles at the second thermal load (i.e. articles placed in the sterilizer for sterilization).

[0157] Additionally, as shown in FIG. 1 the water (provided as steam) provided to the sterilizer 40 is discharged as condensate to a drain 50. In variant examples, the water may be treated (e.g. by water processing equipment such as a filter unit and/or a reverse osmosis unit) and subsequently provided to the water supply vessel 150. However, this would still constitute an open arrangement as defined herein.

[0158] In this example, the water supply vessel 150 has a relatively large volume as a ratio of a storage volume of the pressure vessel 110, as compared with previously-considered steam supply systems. In this particular example, the water supply vessel 150 is sized to receive condensate corresponding to all steam discharged by reducing water stored in the pressure vessel at peak conditions (e.g. a peak storage pressure of 3.101 MPa and a peak liquid level of 0.9 liquid fraction) to a lower limit pressure of 0.101 MPa (8 bar gauge). This corresponds to the water supply vessel having a volume of about 11.5% of the volume of the pressure vessel (corresponding to storage of the subcooled water at 85° C. at atmospheric pressure (0.101 MPa)). In other examples this ratio may be higher or lower, selected depending on an intended mode of use. For example, if the peak storage pressure is higher then the ratio may be larger, and the ratio may be lower if the peak storage pressure is lower. Further, when it is anticipated that a significant proportion of

steam may be discharged to a thermal load in an open arrangement, there may be less water recirculating to the water supply vessel. Previously-considered steam supply systems have a considerably smaller water supply vessel in relation to the size of a pressure vessel (e.g. a boiler or accumulator), because in such systems there is generally a continuous supply of water to the respective pressure vessel. In contrast, as will become clear from the following description, the thermal energy storage and supply systems disclosed herein may be operated to discharge steam from the pressure vessel without necessarily requiring concurrent re-supply of water, which may be subsequently provided when there is an opportunity to recharge the pressure vessel. In particular, a mass of subcooled water provided to the vessel during the depletion period may be relatively low as compared to the mass of steam discharged during the depletion period, for example no more than 50%, or no more than 25%, no more than 10% or no more than 5%. It may be zero in the case that the demand for thermal energy is met without requiring any addition of subcooled water to maintain a sustainable liquid level in the pressure vessel, as will be described in detail below.

[0159] In use, the installation **100** stores and supplies thermal energy to the thermal loads. A significant reserve of thermal energy can be stored in the pressure vessel **110** by heating water contained therein to a storage pressure significantly higher than required by the respective thermal loads. Steam can then be generated (flushed) for supply to the thermal loads by reducing the storage pressure in response to demand. The system is particularly suitable for use with thermal loads which have highly variable thermal energy demands, or which demand a large amount of energy in a relatively short duration (e.g. during peak operational hours of the facility) with significant intervals of low demand therebetween (e.g. overnight).

[0160] FIG. 2 shows an example plot of thermal energy demand (in particular, steam demand) for the industrial plant **10**. The short dashed line labelled “heating demand” corresponds to the steam demand for the heating system **30**, which receives heat via steam supply to the heat exchanger **130**. The units are steam demand in kg/hr and this corresponds to the steam supplied to the heat exchanger **130** in the closed loop arrangement. The long dashed line labelled “sterilizer demand” corresponds to the steam demand for the sterilizer **40**, which receives steam directly from the installation **100** in an open arrangement. The solid line is a summation of the respective demands. As shown in FIG. 2, there is a relatively short window in which a majority of the demand falls, extending from approximately 08:00 (8 o’clock a.m.) to 16:00 (4 o’clock p.m.), largely driven by the sterilizer demand. There is a comparatively low demand in the remaining hours (overnight).

[0161] The installation is operated to build up a significant store of thermal energy in the pressure vessel **110** which can be discharged during a depletion period **302** (indicated above the plot in concordance with the x-axis for time), and which can be subsequently re-accumulated during a recharge period **304**.

[0162] The expression “flash potential” as used herein corresponds to an amount of steam that can be flashed from liquid water within the pressure vessel before the storage pressure reaches a lower limit pressure for sustaining discharge of steam to the thermal load. Where steam is provided to a thermal load in an open arrangement, the asso-

ciated lower limit pressure is a discharge pressure sufficient for sustaining steam flow to be received at the thermal load at a minimum distribution pressure of the thermal load (the discharge pressure typically being about 0.1 MPa greater than the distribution pressure) which may be fixed or variable. If the distribution pressure is variable, it may be controlled independently of the steam installation. In the example installation **100** of FIG. 1, there is a pressure sensor **240** for monitoring the distribution pressure associated with second thermal load, operatively coupled to the controller **200** to provide a distribution pressure signal to the controller **200**, with the controller controlling the respective control valve to discharge steam at a corresponding discharge pressure (e.g. a fixed amount above the distribution pressure). The pressure sensor **240** is shown outside of the industrial plant **10** (i.e. outside of the dashed line), but in a variant example may be replaced by a pressure sensor installed within the industrial plant **10** to monitor a distribution pressure associated with the second thermal load **40** or a steam network of the industrial plant **10** to which it belongs, but remaining operatively coupled to the controller **200**. For example, a distribution pressure for a network of thermal loads may be set at 7 bar gauge (8.101 MPa), with the associated discharge pressure being 8 bar gauge (9.101 MPa).

[0163] Where steam is provided to a thermal load in a closed loop, the lower limit pressure may be variable and dependent on a flow rate required to meet the thermal energy demand of the respective thermal load. For example, the lower limit pressure for a closed loop system may be atmospheric pressure (or below). It may only be limited by the associated saturation temperature and flow rate being high enough to sustain heat transfer to the respective load to meet the thermal energy demand.

[0164] Where steam is provided to thermal loads in both an open arrangement and a closed loop arrangement, the lower limit pressure for assessing the flash potential is the highest of the respective lower limit pressures for the individual loads.

[0165] In this example, the lower limit pressure is the discharge pressure (8 bar gauge, 9.101 MPa) that corresponds to the distribution pressure for the second thermal load (7 bar gauge, 8.101 MPa).

[0166] The expression “liquid level margin” corresponds to an amount of liquid water within the pressure vessel above a lower limit amount for continued operation of the heater upon discharge of the flash potential. It is therefore positive if discharge of the flash potential would leave an amount of liquid water greater than the lower limit amount, and negative if there would be an amount less than the lower limit amount.

[0167] The flash potential at any particular moment therefore corresponds to a maximum demand that may be met. A negative liquid level margin associated with that flash potential indicates that the amount of liquid within the pressure vessel is the limiting factor in being able to meet that demand while continuing to operate the heater, whereas a positive liquid level margin indicates that a demand (e.g. a demand for steam) corresponding to the flash potential can be met while continuing to operate the heater.

[0168] The flash potential can be readily calculated by the person skilled in the art. It is proportional to the amount of liquid water in the pressure vessel, and the difference between the enthalpies of liquid water at the storage pressure

and lower limit pressure (which approximately scales with the pressure difference). It is inversely proportional to the enthalpy of vaporization at the lower limit pressure. Similarly, the liquid level margin can be readily calculated based on the quality of a known saturated mass of water at a given pressure. Nevertheless, both quantities can be indirectly monitored or can be indirectly controlled by monitoring associated quantities. For example and as described in detail below with reference to FIGS. 7a-7e, a flash potential and liquid level margin for a given storage pressure and lower limit pressure can be determined based on the liquid level. Operation of the installation by reference to the liquid level (e.g. by evaluating a criterion based on the liquid level) can therefore correspond to operating the installation to control a flash potential and/or liquid level margin as described herein. Except for explicit references to a calculation of a flash potential, liquid level margin or associated quantity, references herein to control based on a flash potential and/or based on a liquid level margin may be implemented without direct calculation of the flash potential and/or liquid level margin, with suitable implementations of the control being based on a monitored liquid level and storage pressure, for example by reference to a database of level thresholds correlated to storage pressure.

[0169] Storage pressure and liquid level are important quantities in the operation of the installation to store and dispense thermal energy from the pressure vessel.

[0170] The principle variables which affect conditions within the pressure vessel are (i) the discharge of steam by flashing, (ii) the addition of subcooled water and (iii) heating using the heating element, each of which may be performed in isolation or in combination with each other. (i) Flashing steam from the pressure vessel is achieved by lowering the storage pressure, which in isolation causes the liquid level in the vessel to fall. (ii) Adding subcooled water into the vessel in isolation causes the storage pressure to drop and the liquid level to rise. (iii) Heating in isolation causes the storage pressure to rise and the liquid level to rise.

[0171] References herein to the supply of subcooled water refer to the water being subcooled relative to the saturation temperature at the storage pressure of the pressure vessel. As described elsewhere herein, an inlet flow of water to the pressure vessel may be heated to saturation temperature as it flows into the pressure vessel, for example in a deaerator 120. Nevertheless, such heating utilizes heat energy from within the pressure vessel, and so it is considered appropriate to refer to the state of the inlet as subcooled by reference to its condition prior to entry to the pressure vessel, as this aids an understanding of the energy balance and effect of supplying the (subcooled) water to the pressure vessel.

[0172] FIGS. 3a and 3b are plots of operating conditions during an example 24 hour operating cycle of the installation 100 including a depletion period and a recharge period. FIG. 3a shows trends of storage pressure and liquid level (within the pressure vessel). FIG. 3b shows trends of flash potential and liquid level margin.

[0173] FIGS. 3a and 3b are based on a simplified set of operating variables, including a simplified profile of thermal energy demand relative to that shown in FIG. 2. The example operating variables include a constant maximum steam demand of approximately 330 kg/hour for the duration of a depletion period lasting approximately 6.5 hours, and no demand for a recharge period which lasts the remainder of the 24 hour cycle.

[0174] No power is available for use by the heater in the depletion period, but power for heating is available throughout the recharge period at a constant rate of 120 kW. Such a scenario may arise in practice, for example if electric power is obtained from a local power source (such as a renewable energy power source) and is utilised for other purposes during the depletion period (e.g. during the day), and is only available for heating during the recharge period (e.g. during the night).

[0175] Also for simplicity, the plots of FIGS. 3 and 4 are shown with time starting at the start of the depletion period. The initial conditions of the pressure vessel upon commencement of the 24 hour cycle (i.e. upon commencement of the depletion period) are a peak storage pressure of 30 bar gauge (3101 kPa), and a peak liquid water fraction of 90%. A lower limit pressure for sustaining discharge of steam to the thermal load is 8 bar gauge (9101 kPa), which corresponds to the distribution pressure for the thermal load (e.g. the distribution pressure being 7 bar gauge (8101 kPa) with an offset of 1 bar (100 kPa)). A lower limit liquid level for continued operation of the heater corresponds to a liquid fraction of 0.5. The vessel size is approximately 22.5 m³.

[0176] In this simplified example, the steam demand is selected so that the flash potential is completely exhausted at the end of the depletion period, and so the plots of FIGS. 3a and 3b do not represent a failure to meet the steam demand which ends the depletion period, despite the flash potential reducing to zero.

[0177] As shown in FIGS. 3a and 3b, during the depletion period the storage pressure progressively reduces from the peak storage pressure of 30 bar gauge (3101 kPa) to flash sufficient steam to meet the (constant) steam demand, thereby reducing the flash potential. The liquid level also decreases as water is discharged (in the form of steam) from the pressure vessel. As the flash potential is always sufficient to meet the demand and no condensate is added during the depletion period, the liquid level margin remains constant during the depletion period. The flash potential progressively reduces to reflect the progressive discharge of steam.

[0178] At the end of the depletion period, operation transitions to a recharge period lasting the remainder of the 24 hour cycle, and in which there is no demand.

[0179] During the recharge period, the installation is operated to both (i) re-supply the pressure vessel with subcooled water, to replace that which has been discharged (in the form of steam), and (ii) add heat to re-pressurise the pressure vessel to the peak storage pressure. In particular, water is added to reach a target (or peak) mass of water corresponding to the peak liquid fraction at the peak storage pressure, whereas heat is added to reach the peak storage pressure given the target mass of water.

[0180] The supply of subcooled water and of heat during the recharge period may be managed or phased in any suitable manner.

[0181] In the specific example of FIGS. 3a and 3b, heat is provided at a constant rate (120 kW), whereas subcooled water is provided according to a front-loaded profile. By front-loading the supply of subcooled water, the water in the pressure vessel may be maintained at a relatively low pressure and temperature (and a relatively high liquid level) during an early portion of the recharge period, only reach higher pressures and temperatures during a final portion of the recharge period, thereby minimizing thermal losses through walls of the pressure vessel for as long as possible.

[0182] In this particular example, the supply of subcooled water is controlled based on the flash potential and the liquid level margin as will be described below. The flash potential is zero at the beginning of the recharge period as the storage pressure has been reduced to the lower limit pressure for sustaining steam discharge to the thermal load, and as such there is no capacity to provide additional steam in the event it were to be demanded.

[0183] For an initial flash-priority portion of the recharge period, heat is provided without corresponding supply of subcooled water to build the flash potential up to a minimum recharge flash potential (a predetermined value of 500 kg, in this example) that may be utilized if required. The liquid level margin reduces during this initial flash-priority portion of the recharge period. Supplying subcooled water during this initial portion would slow the rate of pressure increase (or even decrease pressure), slowing the build-up of the flash potential.

[0184] In a subsequent water-priority portion of the recharge period (commencing at approximately hour 9), subcooled water is provided at a rate such that the minimum recharge flash potential is maintained while the liquid level margin increases. This water-priority mode of operation is maintained until the mass of water reaches the target (or peak) mass.

[0185] Upon reaching the target mass of water (at approximately hour 11), the installation is operated in a second flash-priority portion of the recharge period, in which heat is provided without corresponding supply of subcooled water, until the target peak storage pressure is reached. In this example, the target peak storage pressure is reached at approximately hour 20, with no further water or heat being added during a final period of the 24 hour cycle of approximately four hours. Alternatively, given the predicted demand and predicted heating power availability, the water supply and/or heating profiles may be varied to delay the second flash-priority period (for example by providing no heat or subcooled water for a four hour portion between the water-priority portion and the second flash-priority portion). In other examples of use, particularly those having a steam demand during the recharge period, further subcooled water may be provided during the second flash-priority period in order to target or maintain the target (peak) mass of water in the pressure vessel.

[0186] FIGS. 4a and 4b are plots of operating conditions for a second example of operating the installation including a depletion period and a recharge period, showing the same quantities as in FIGS. 3a and 3b. In the second example, the operating variables are the same as in the second example except for that power for heating is available throughout both the depletion period and the reheat period (at the same power of 120 kW), and the steam demand is significantly higher at 500 kg/hour. The lower limit liquid level is also higher and corresponds to a liquid fraction of 0.67, to simulate the impact of a higher required liquid level.

[0187] The profiles of FIGS. 4a and 4b are similar to those of FIGS. 3a and 3b in that the pressure and liquid level reduce during the depletion period. The significantly higher steam demand rate is offset by the heating power available during the depletion period. However, in this example the liquid level margin reaches zero in a final portion of the depletion period, meaning that if the flash potential were to be discharged by lowering the storage pressure to the lower limit pressure, the liquid level would be at the lower limit

liquid level for continued operation of the heater. Subcooled water is added to the pressure vessel during this final portion of the depletion period to maintain the liquid level margin, and this accelerates the rate at which the storage pressure and flash potential reduce, but the flash potential is still not exhausted. The liquid level does not reach the lower limit during the depletion period, in particular because the full flash potential is not demanded (as shown in FIG. 4a).

[0188] In the recharge period, there is a brief initial flash-priority portion in which the flash potential is raised to a minimum recharge flash potential (250 kg, in this example). A water-priority portion then follows in which the minimum recharge flash potential is maintained while the liquid level margin increases, until the target mass of water is reached. There then follows a second water-priority portion in which heat is added without corresponding supply of subcooled water, until the target peak storage pressure is reached. Finally, there is a period towards the end of the 24 hour cycle in which the monitored quantities remain constant.

[0189] Other methods for phasing the supply of subcooled water and heat may be applied. For example, at the start of a recharge period (or at any point during the recharge period), the controller may determine an additional mass of subcooled water to be added to a current mass of water within the pressure vessel to reach the target mass of water. The current mass of water may be determined based on the liquid level as derived from the level signal, and based on the storage pressure as derived from the storage pressure signal. The controller may determine a total amount of heat to be provided to the water (current and additional) during the recharge period to reach the peak storage pressure. The controller may control a rate at which the subcooled water is supplied to the pressure vessel so that it is proportional to the heating power, thereby resulting in a continuous gradual rise in both the storage pressure and liquid level (/liquid fraction) during the recharge period.

[0190] Since the pressure vessel is configured and operated to provide a large reserve of thermal energy (e.g. a large flash potential) relative to a maximum expected rate of steam demand, in use the flow rate of steam out of the vessel is relatively low compared to a maximum which the pressure vessel could potentially accommodate. This has advantages relating to the dryness of the steam. In particular, for on-demand steam generation systems or short-term steam accumulator systems, it is conventional to determine whether the maximum expected rate at which steam is released from the surface of liquid water is sufficiently low to inhibit entraining liquid water. Empirical test work conducted by the applicant and reported before the priority date shows that the rate at which dry steam can be released from the surface of water is a function of pressure, with a working approximation that the maximum steam release rate (MSR) without steam entrainment ($\text{kg/m}^2\text{h}$) is equal to the absolute pressure in bar multiplied by 220 (or 22 times the absolute pressure in MPa). It is conventional practice, for example in the design of a steam accumulator, to seek to reduce the size of the respective vessel to reduce cost and installed size, thereby resulting in a maximum expected (or peak) steam release rate being relatively high compared to the maximum steam release rate (MSR), for example between 25% and 90%. In contrast, the thermal energy storage and supply installation and methods as described herein will tend to result in far lower steam release rates. In the particular

example described above with references to FIGS. 3a and 3b, the maximum expected or peak steam release rate for a suitably sized vessel equates to approximately 0.7% of the MSR for the peak storage pressure and 2.3% of the MSR for the lower limit pressure, even when both are conservatively assessed assuming a 90% fill. For an 80% fill the values are approximately 0.6% and 2% respectively. Accordingly, the risk of entraining liquid water in the discharged steam is low.

[0191] FIGS. 5-7 are flow diagrams of methods for operating a thermal storage and supply installation by controlling respective control variables of such an installation. The methods will be described by reference to the example thermal storage and supply installation 100 of FIG. 1, and with reference to the profiles of operating conditions shown in FIGS. 3a and 3b.

[0192] FIG. 5 shows a method 500 executed by the controller 200 for controlling the discharge of steam from the pressure vessel 100 to a thermal load, and will be described by way of example with reference to a demand for thermal energy at the first plant thermal load 30, although it will be appreciated that the method applies equally to demand at another thermal load or a cumulative demand across multiple loads. The steps of the method as described with reference to the “blocks” of FIG. 5 are continuously or periodically repeated during operation of the installation.

[0193] In block 502, the controller determines whether there is a demand based on the demand signal received from the first plant thermal load 30. If there is no demand then the method goes to block 510 and the respective discharge control valve 216 is closed or remains closed.

[0194] If there is a demand, then the method goes to block 504 and the controller determines whether the liquid level is above the lower liquid level limit, for example based on the level signal received from the level sensor. If the liquid level is not above the lower liquid level limit, the method goes to block 510 and the respective discharge control valve 216 is closed or remains closed. This state may persist until the liquid level is raised, for example by the addition of heat and/or the addition of subcooled water to the pressure vessel as described above with reference to FIGS. 3-4.

[0195] If the liquid level is above the liquid level limit, the method moves to block 520 and the controller controls the respective first discharge control valve 216 to open to lower the storage pressure in the pressure vessel 110, to thereby cause liquid water to flash and be discharged through the outlet 116, through the first discharge control valve 216 and along the distribution line 102 to the heat exchanger 130. As described elsewhere herein, the controller may control the first discharge control valve 216 based on a monitored parameter corresponding to the supply of steam to the respective thermal load (e.g. in a feedback loop), for example based on the first flow rate signal provided by the first flowmeter 218.

[0196] In block 522, the controller may determine whether the demand is being met. For example, the controller may determine whether a flow rate of steam along the respective distribution line 102 is sufficient to meet the demand based on the first flow rate signal received from the respective first flowmeter 218. The demand may not be met if there is insufficient flash potential of water within the pressure vessel to meet the demand. If the controller determines that the demand is not being met, the controller may determine to close the discharge control valve 216 until the flash potential increases, for example by further heating and/or

water supply to the pressure vessel. Responsive to determining that the demand is not being met, the controller may control additional heat to be provided, for example using electrical power from a secondary power source (e.g. from an electrical grid connection as opposed to a local power source (such as a microgrid) delivering renewable energy such as solar or wind power). Whether or not the demand is being met, the method is repeated by return to block 502 throughout operation of the installation 100.

[0197] FIG. 6 shows a method 600 of controlling the heater when the heater is powered by a local power source, which may be performed concurrently with and independently of the method 500 of FIG. 5. As above, it will be described by reference to the example installation 100 of FIG. 1. This example method does not refer to the optional supply of electrical power from a secondary power source such as an electrical grid connection, which is described elsewhere herein.

[0198] The thermal energy storage and supply installation 100 confers particular advantages when powered by a local power source, such as a microgrid. Local power sources and in particular renewable power sources generally do not provide power at the same consistency and reliability as a non-local electrical grid that distributes power from a large number of different renewable and non-renewable power stations, for known reasons (e.g. variable amount of wind or solar energy). The thermal energy storage and supply installation 100 provides a large reserve for converting electrical energy into thermal energy for storage as it becomes available, and subsequently releasing that thermal energy on demand. The inventors have found it to be more efficient to do so than to store locally generated energy in an electrical battery for subsequent use in an electric boiler on demand. Further, storing energy as thermal energy as described herein has advantages over electrical battery storage as it does not require vast quantities of the chemicals and materials associated with electrical battery technology (many of which are rare or environmentally damaging to mine), such as lithium.

[0199] The rates at which energy is stored and discharged by the installation are largely uncoupled. Accordingly, a large reserve of thermal energy can be built up in the pressure vessel by heating liquid water within the pressure vessel as and when it becomes available. Generally (i.e. across most of an operating map of pressure vessel conditions), heat can be supplied without requiring corresponding supply of subcooled water to the vessel. The principal exceptions are scenarios which are easily avoided, namely: (i) when the pressure vessel is at the peak storage pressure but the mass of water in the pressure vessel is lower than the target peak mass of water; and (ii) when the liquid level is below a lower limit level for continuing operation of the heater, for example a lower limit level corresponding to the uppermost position of the heater or its heating element(s) within the pressure vessel. In the case of (i), additional subcooled water should be provided to increase the mass of stored water and reduce the pressure. In the case of (ii), additional water should be provided to increase the liquid level, as in the circumstances the heater cannot be operated to raise the water level.

[0200] In block 602, the controller determines whether power for heating is available from the local power source. The determination may be a determination of a power currently being generated by the local power source and/or a power currently available from the local power source

considering other loads (e.g. the generated power minus power provided to other electrical loads in the installation 100 and/or the industrial plant). If power is not available, the method moves to block 610 and the controller does not operate the heater to heat water within the pressure vessel. If power is available, the method moves to block 604.

[0201] In block 604, the controller determines whether the storage pressure is already at the peak storage pressure (e.g. based on the storage pressure signal received at the controller). If it is, then the method moves to block 610 for not heating, as described above. If the storage pressure is less than the peak storage pressure, then heat is provided.

[0202] In block 606, the controller determines if the liquid level is within a range for heating (e.g. based on the level signal received at the controller). In this example, the determination corresponds to (i) determining whether the liquid level is above the lower limit level for continued operation of the heater, which may correspond to the uppermost position of the heater or its heating element(s) within the pressure vessel as mentioned above and (ii) determining that the liquid level is not above an upper limit level corresponding to the target peak mass of water in the pressure vessel. The upper limit level may be pressure-dependent. In particular, heating the vessel in isolation (i.e. in the absence of steam discharge or water supply) causes the pressure to rise and the liquid level to increase. Accordingly, if the vessel is at a target peak liquid fraction (e.g. 90%) at a storage pressure lower than the target peak storage pressure, then further heating to reach the target peak storage pressure would cause the liquid level to rise further. The controller may control the supply of water to the pressure vessel to avoid such a scenario arising, but nevertheless it may optionally be checked for in block 606.

[0203] If the liquid level is not within the heating range, then the method moves to block 610 as described above and no heat is provided. However, if the liquid level is within the heating range, then the method moves to block 620 and heat is provided at a rate corresponding to the power availability.

[0204] Whether or not heat is provided, the method is repeated by return to block 602 throughout operation of the installation 100.

[0205] The heater may have a rating (i.e. a maximum power output) considerably lower than that of heaters or burners that may be provided in alternative steam supply systems. Whereas such alternative steam supply systems may size the heater based on a peak steam supply rate, a heater in accordance with the present disclosure may be sized based on the power supply and a total amount of heat that may be required during an operating cycle, such as a day. Considering the mode of operation of the installation (i.e. to build up a significant reserve of thermal energy over a long time period, which can be discharged at a rate far exceeding the simultaneous availability of power for heating), the inventors have provided a system which can satisfy a relatively high cumulative thermal energy demand and a relatively high rate of demand, with a relatively low heating power. The heater itself may have a relatively low rating, or it may be coupled to a local power source having a relative low power output, or the controller may limit a power output of the heater.

[0206] To quantify the relatively low heating power as compared with the relatively high capacity of the system to meet a demand, the present disclosure defines the following parameters:

[0207] An average depletion power is defined as a cumulative enthalpy of steam discharged from the pressure vessel during the depletion period divided by a duration of the depletion period;

[0208] A maximum depletion power is defined as a maximum enthalpy of steam discharged within the depletion period during any minimum power evaluation period of one minute, divided by the minimum power evaluation period. It therefore typically corresponds to a peak rate of steam discharge, taking into account that steam discharged at a lower pressure may have a lower enthalpy than steam discharged at a higher pressure.

[0209] An average reheat power is defined as accumulative energy provided to liquid water by the heater during the recharge period, divided by a duration of the recharge period;

[0210] A maximum reheat power is defined as a maximum power at which energy is provided to the liquid water by the heater within the recharge period.

[0211] According to the present disclosure, the average reheat power during the recharge period may be significantly lower than the average depletion power, for example no more than 50% of the average depletion power, corresponding to the configuration of the system to use a relatively low power source or low rating heater to meet a relatively high energy demand during a limited period. In the specific example of FIG. 3, the average depletion power is 0.26 MW whereas the average reheat power is 0.12 MW. This ratio would be lower if heating were to be provided during the depletion period.

[0212] According to the present disclosure, the average reheat power may be no more than 50% of the maximum depletion power. In the specific example of FIG. 3, the maximum depletion power corresponds to discharge of steam when the pressure vessel is at the peak storage pressure, at the constant demand rate of approximately 330 kg/hr. This corresponds to a maximum depletion power of approximately 0.26 MW.

[0213] According to the present disclosure, the maximum reheat power may be no more than 50% of the maximum depletion power. The corresponding values for the specific example of FIG. 3 are provided above.

[0214] According to the present disclosure, a dimensional ratio of (i) the cumulative enthalpy of steam discharged from the pressure vessel during the depletion period and (ii) the average reheat power of the heater during the recharge period is at least 25000 seconds. This corresponds to the reheat power being relatively low as compared with the capacity of the system to meet a thermal load, which differs in the present invention as compared with previously considered steam supply systems for comparable applications. Although the units are seconds, this does not correspond to a time period of the recharge period. In the specific example of FIG. 3, the dimensional ratio is approximately 50000 seconds.

[0215] The rating of the heater may be selected in part based on the variability of the supply. For example, where electrical power can reliably be drawn from the power supply, the rating of the heater may be equal to the total expected heat energy to be provided by the heater over a cycle, divided by the duration of the cycle. However, for a power supply which has a variable output (such as wind or solar), a suitable rating may be determined by evaluating a

predicted power output profile of the local power source. It may be that a local power source can generate a peak power above the rating of the heater. As mentioned elsewhere herein, supplementary power may be provided (e.g. from a non-local electrical grid connection) in the event that a local power source is unable to meet a power output requirement to achieve a target storage pressure and/or a target water mass (which may be a target peak storage pressure and a target peak water mass).

[0216] Before describing a method of controlling the supply of subcooled water with reference to FIG. 8, it is helpful to illustrate of the expressions “flash potential” and “liquid level margin” as used herein, as shown in FIGS. 7a-7e.

[0217] FIGS. 7a-7e show a column-shaped vessel 710 representative of the pressure vessel 110 of FIG. 1. The line L0 represents a lower limit liquid level of the vessel 710 for continued operation of the heater (i.e. corresponding to a height of the respective heater or at least heating elements of the heater). The line 714 represents the liquid level of water in the vessel 710. As noted elsewhere herein, the liquid level may be expressed as a height of the interface between liquid water and steam, or the liquid fraction within the vessel. In each drawing, the current storage pressure of water is the same (and below a target peak storage pressure for the vessel), but the mass of water and therefore the liquid level is different.

[0218] In FIG. 7a, the liquid level 714 is below the lower limit liquid level L0, such that the heater cannot be operated. This condition may only occur in operation when the installation is intentionally operated to exhaust a flash potential even while taking the liquid level below the lower limit, thereby preventing further heat input during this operation until the liquid level is increased by the supply of further water into the vessel.

[0219] In each of FIGS. 7b-7d there are two side-by-side representations of the vessel 710. The left-hand representation shows a current liquid level 714 at the current storage pressure, whereas the right-hand representation shows what the liquid level 716 would be after discharge of the flash potential.

[0220] In FIG. 7c, the liquid level 714 is at a zero-margin level L1 corresponding to the liquid level margin being zero (i.e. the liquid level upon discharge of the flash potential being equal to the lower limit liquid level L0, as shown in the right-hand representation of the vessel 710).

[0221] In FIG. 7b, the liquid level 714 is at an intermediate point between level L0 (lower limit liquid level) and L1 (zero-margin liquid level), such that the liquid level margin is negative.

[0222] In FIG. 7e, the left-hand representation of the vessel 710 shows a current liquid level 714 at the current storage pressure, whereas the right-hand representation of the vessel 710 shows the liquid level 716 of the same mass of water if the storage pressure of the water were to be raised to a target peak storage pressure. In FIG. 7e, the current liquid level 714 is at a peak fill level L2 corresponding to the liquid level 716 at a (higher) target peak storage pressure being a target peak liquid level, for example 90%.

[0223] In FIG. 7d, the liquid level 714 is at an intermediate level between levels L1 (the zero-margin level) and L2 (the peak fill liquid level).

[0224] As may be appreciated, the lower limit liquid level is the same at every storage pressure, whereas the zero-

margin liquid level L1 and the peak fill liquid level L2 vary depending on storage pressure.

[0225] In the absence of further heat input while a flash potential is being exhausted, the optimum liquid level at any particular storage pressure would be the zero-margin liquid level L1. This represents depressurization of the pressure vessel to the lower limit pressure, while reaching but not going below the lower limit liquid level L0.

[0226] When the liquid level 714 is between L0 and L1, and in the absence of heat input while a flash potential is being exhausted, only part of the flash potential (a “useful flash potential”) may be discharged before the liquid level reaches the lower limit liquid level L0. Generally, the flash potential (including a useful flash potential) increases in response to heat input, since both the liquid level and the enthalpy of liquid water increases. Generally, the flash potential decreases in response to subcooled water input, since this reduces the temperature and enthalpy of the liquid water, albeit raising the total amount of liquid mass. However, when the liquid level 714 is between L0 and L1, the useful flash potential tends to increase in response to subcooled water input, because the amount of water serves as the limiting factor in permitting steam to be flashed to vapour and discharged. Accordingly, the addition of water permits more of the flash potential to be utilized, even though it reduces the total flash potential.

[0227] When the liquid level 714 is between L1 and L2, the flash potential (now equal to the useful flash potential) would increase in response to heat input. The flash potential (again corresponding to the useful flash potential) would decrease in response to subcooled water input, because the liquid level does not serve as the limiting factor in permitting discharge of the flash potential, whereas the addition of subcooled water lowers the pressure and the enthalpy of liquid water within the pressure vessel.

[0228] Accordingly, a useful flash potential is best conserved or increased by: adding subcooled water when the liquid water level is between L0 and L1 for a given pressure, and adding heat without adding subcooled water when the liquid water level is between L1 and L2 for a given pressure.

[0229] Of course, it is also necessary to add subcooled water to the pressure vessel when the liquid water level is between L1 and L2. However, when to add such water may depend on the operating mode of the installation.

[0230] The above discussion is applicable to a flash potential as an instantaneous flash potential—i.e. the amount of steam that may be flashed by lowering the pressure to the lower limit pressure, based on the current mass of water and storage pressure within the pressure vessel. However, the above discussion is also applicable to a predicted flash potential, predicted liquid level margin, and predicted useful flash potential, determined based on a predicted demand and/or a predicted power output of the heater.

[0231] To illustrate how the predicted flash potential may differ from an instantaneous flash potential, a scenario can be considered in which the liquid level is 0.75, the instantaneous flash potential is X kg (e.g. 500 kg) and there is a positive liquid level margin Y (e.g. a liquid fraction of 0.1 above a lower limit liquid level of 0.5). If the predicted demand corresponds to 250 kg/hr of steam discharge per hour for two hours, then in the absence of additional heating and supply of subcooled water, the flash potential will be exhausted at the end of the two hour period and the liquid level will be at the lower limit liquid level L0. However, if

there is a predicted power output of the heater of 100 kW over the same two hour period, then a lower reduction of pressure would be required over the two hour period in order to discharge the same amount of steam (because the heat input also vaporizes liquid water), and there would still be a positive flash potential at the end of the two hour period. However, it may be that the liquid margin becomes negative, limiting the useful flash potential.

[0232] The flash potential, liquid margin and/or the useful flash potential may be evaluated over a time period based on a predicted demand and a predicted output of the heater, as will be described in further detail below. This may be particularly useful for controlling the supply of subcooled water during a depletion period as will be described below. In the context of the levels L0, L1, L2 as discussed above with respect to FIG. 7, these predicted quantities may particularly affect the zero-margin level L1 (the level L0 is relevant only for the current liquid level, whereas the level L2 is most relevant for recharging periods).

[0233] Continuing the above specific example scenario, the zero-margin liquid level L1 based on an instantaneous flash potential and liquid level margin may be 0.65, corresponding to a reduction of the liquid level by 0.15 to reach the lower limit liquid level of 0.5. However, the zero-margin liquid level L1 based on a predicted flash potential and predicted liquid level margin may be lower, for example 0.6, because a lower reduction in storage pressure may be required to output the same amount of steam during the two hour period when the heater is providing additional energy into the pressure vessel. Further, with the higher storage pressure, the liquid level may drop at a slower rate.

[0234] A similar analysis could be applied to determine a peak fill liquid level L2 based on a predicted demand and predicted power output. However, as compared with an analysis based on the current conditions of the pressure vessel, it may be that the effect of using predicted quantities may only be a determination to add more water into the pressure vessel above the target peak mass of water, in anticipation of some of that water being discharged to meet the predicted demand. While there may be efficiency advantages to doing this (in particular to lower the temperature of the pressure vessel to reduce thermal loss), a practical control implementation may reflect the target peak mass of water as an upper limit. For example, if the pressure vessel was filled with a mass of water greater than the target peak mass at a relatively lower pressure, then in the event that a predicted demand does not come about, then the liquid level would exceed the associated target peak liquid level at the target peak storage pressure.

[0235] FIG. 8 shows a method 800 of controlling the supply of subcooled water to a pressure vessel, which may be performed concurrently with and independently of the methods 500, 600 of FIGS. 5 and 6. As above, it will be described by reference to the example installation 100 of FIG. 1.

[0236] As discussed elsewhere herein, in isolation (i.e. aside from any heating by the heater and/or discharge of steam), supplying subcooled water into the pressure vessel reduces the storage pressure of water in the pressure vessel. Accordingly, while operation of the installation 100 to discharge steam relies on there being an adequate mass of water within the pressure vessel, in many parts of the operating map of the installation, the act of resupplying

water into the pressure vessel tends to reduce the potential of the pressure vessel to discharge steam (i.e. the flash potential).

[0237] The inventors have considered that the timing and rate of water supply can be controlled depending on an objective for operation of the installation.

[0238] In block 802 of the method, the controller 200 determines whether the liquid level is less than the lower limit liquid level L0 for continued operation of the heater. The determination may be based on a level signal received from the level sensor 214. If the liquid level is below the lower limit liquid level L0, the method continues to block 820.

[0239] In block 820, an amount of subcooled liquid to supply is determined. If the installation is being operated in an extended depletion mode in which the liquid level is permitted to extend below the lower limit liquid level L0 in order to discharge the full flash potential, then the controller determines not to add subcooled water. In all other modes (i.e. in a heated depletion mode in which the controller prevents discharge of steam that would cause the liquid level to fall below the lower limit liquid level for operation of the heater, or in a recharge mode), the controller determines to add subcooled water.

[0240] The amount of subcooled water to supply may be determined (block 820) based on both the liquid level and the storage pressure. For example, the controller may calculate an amount of subcooled water required to reach the lower limit level, or may look up an amount from a database of predetermined values correlated to liquid levels and storage pressures. The amount of subcooled water can be specified in any suitable way, for example as a mass of water, a duration of operating an inlet pump and/or a flow rate.

[0241] In block 822, a flow rate for supplying the subcooled water is determined. In the context of restoring the liquid level to the lower limit liquid level L0, the heater is required to be inactive and so the flow rate is determined without reference to the rate of heating. If the controller determines that the subcooled water is to be deaerated, the flow rate may be determined based on predetermined inlet flow rate associated with operation of the deaerator, which may correspond to an inlet flow rate at which the deaerator is able to raise the temperature of the inlet flow to the saturation temperature. Otherwise, the flow rate may be determined by an inlet pump, for example as an optimal or max.

[0242] In block 824, the subcooled water is caused to be supplied up to the amount and/or at the flow rate as determined in blocks 820, 822.

[0243] While both an amount of subcooled water to add and a supply rate (block 822) are determined in this particular example, in variant examples the controller may proceed directly to block 824 to commence supplying subcooled condensate, for example at a predetermined rate. Repetition of the method (as described above) may provide suitable control for stopping the supply.

[0244] The method continues to block 802 to repeat the method. The method may be repeated before a determined amount in the last iteration is provided, such that any requirement for supplying subcooled water to the pressure vessel is repeatedly determined.

[0245] If the liquid level is determined to be at or above the lower limit liquid level L0 at block 802, the method

continues to block **804**. At block **804**, it is determined whether the liquid level margin is negative. The liquid level margin corresponds to the amount by which the liquid level in the pressure vessel would be above or below the lower limit liquid level L_0 upon discharge of the flash potential.

[0246] Determining that the liquid level margin is negative corresponds to determining that the amount of water in the pressure vessel is currently the limiting factor on the ability of the pressure vessel to discharge steam. The inventors have found that the useful flash potential (as defined above) can be increased by adding subcooled water to the pressure vessel. While this reduces the storage pressure and the flash potential (to any liquid level), it increases the useable flash potential. If the liquid level margin is negative then the method proceeds to block **820**.

[0247] In blocks **820-824**, an amount of subcooled water to supply is determined (block **820**) as described above; a flow rate for the supply is determined (block **822**); and the controller causes the determined amount to be supplied; before repeating the method. As in the context of restoring the liquid level to the lower limit liquid level L_0 , the amount of subcooled water to supply can be calculated, or may be looked up from a database of predetermined values correlated to liquid levels and storage pressures. As in the context of restoring the liquid level to the lower limit liquid level L_0 , the flow rate can be determined without reference to a rate of heating, since the advantage of increasing the useful flash potential is achieved based on the mass of water being supplied, irrespective of the rate of supply.

[0248] A determination at block **804** that the liquid level margin is zero or positive corresponds to determining that the amount of water in the pressure vessel is not currently the limiting factor on the ability of the pressure vessel to discharge steam. The determination may be based on a predicted flash demand and predicted liquid level margin (themselves determined based on current conditions of the pressure vessel (storage pressure, liquid level) and a predicted demand and/or a predicted power output of the heater as described elsewhere herein.

[0249] When it is determined that the liquid level margin is zero or positive, the method proceeds to block **806**.

[0250] At this stage, whether or not subcooled water is supplied to the pressure vessel may depend on the particular application for which the installation is operated, and/or a predicted demand.

[0251] When operating the installation to meet a substantial or uncertain demand, it could be considered counterproductive to add subcooled water since this would reduce the flash potential (and also the useful flash potential). It may also be considered unnecessary, since the liquid level margin being zero or positive corresponds to complete exhaustion of the flash potential leaving sufficient liquid water in the pressure vessel for continued operation of the heater. When the liquid level margin is a predicted liquid level margin over a time period and is zero or positive, this corresponds to the complete exhaustion of the flash potential over the time period leaving sufficient liquid water in the pressure vessel for continued operation of the heater.

[0252] Accordingly, when operating the installation to meet a substantial or uncertain demand, the inventors have determined that the addition of subcooled water can be delayed to avoid unnecessarily reducing a useful flash potential.

[0253] In contrast, in order to recharge the vessel to target peak conditions (i.e. to a target peak storage pressure, with a target peak mass of water) it is necessary to re-supply it with both additional subcooled water to reach the target mass, and heat to reach the target pressure.

[0254] The inventors have determined that the installation may have different operational modes in order to meet these dual requirements, as described by way of example with reference to the method **800** of FIG. **8**.

[0255] In block **806**, the controller determines whether to operate the installation in a depletion mode or a recharge mode. The depletion mode corresponds to operation of the installation during a depletion period to meet a thermal demand that generally depletes the thermal energy stored in the pressure vessel over a sustained time period. A depletion period may be described as a period in which an enthalpy output (or loss) corresponding to the discharged steam is greater than an energy input to the pressure vessel, such that the storage pressure reduces by a significant amount such as at least 1 MPa from the peak storage pressure. The recharge mode corresponds to operation of the installation during a recharge period during which thermal energy generally accumulates in the pressure vessel over a sustained time period. A recharge period may be described as a period in which an energy input to the pressure vessel by heating is greater than an enthalpy output (or loss) corresponding to steam discharge to meet the thermal energy demand whereby the storage pressure increases by at least 1 MPa to the peak storage pressure (e.g. at least 2 MPa). As discussed elsewhere herein, in variant examples in which the peak storage pressure may be lower (e.g. at least 0.5 MPa, or at least 1 MPa, or at least 2 MPa), the depletion period and recharge period may be defined commensurately differently, for example by reference to a depletion pressure difference and/or a recharge pressure difference which is at least 50% of the respective peak storage pressure.

[0256] The controller may determine whether to operate the installation in the depletion mode or the recharge mode in any suitable way, examples of which include (i) a time-based schedule for operation in the respective modes (e.g. 09:00-17:00 in the depletion mode, 17:00-09:00 in the recharge mode); (ii) by reference to a predicted demand and/or predicted power output of the heater over a time period, such as in the next 6 hours, or the next 12 hours, or (iii) by reference to a prediction as to which of a depleted condition or target peak conditions the pressure vessel is predicted to reach first, based on a predicted demand and/or predicted power output of the heater. By way of example, in the case of (iii), a depleted condition may correspond to exhaustion on a flash potential to leave a liquid level margin of zero, or to a useful flash potential being reduced to a minimum depletion potential, such as 500 kg.

[0257] If it is determined to operate the installation in the depletion mode at block **806**, the method proceeds to block **810**. At block **810**, the installation is operated to not add subcooled liquid to the pressure vessel, before returning to block **802** to repeat the method. By not adding subcooled water, the flash potential (a useful flash potential) may be maintained as high as possible during the depletion period, while continuing to meet the steam demand.

[0258] If it is determined to operate the installation in the recharge mode at block **806**, the method proceeds to block **808**.

[0259] In block **808**, it is determined whether to operate in the recharge mode to prioritise flash potential (“flash priority”), or to prioritise water mass accumulation in the pressure vessel (water priority). The inventors have determined that an optimum balance between conserving and accumulating an instant capability to meet a demand (i.e. a flash potential) and storing additional water to build a yet greater demand at a future time can be balanced by prioritizing storing additional water only once a minimum flash potential has been established.

[0260] Considering that the liquid level margin must be positive at block **808** of the method **800**, the flash potential and useful flash potential are equal and correspond to a demand that can be met without additional heat input to the vessel. In this example, the controller determines whether the flash potential is below a minimum recharge potential, which may be a predetermined minimum potential (e.g. generally applicable or alternatively determined from a database of predetermined minimum potentials correlated by an operating variable such as time of day, day of week), or may be determined based on a predicted demand and/or predicted power output of the heater over a time period, as described above. For example, the minimum recharge potential may correspond to a flash potential of 500 kg.

[0261] The controller may determine whether the flash potential is below the minimum recharge potential in any suitable way, including based on the liquid level signal received at the controller (e.g. by reference to a database of liquid level signals corresponding to a minimum recharge potential correlated by storage pressure and optionally the minimum recharge potential (if variable)); or by calculation of the flash potential and comparison with the minimum recharge potential.

[0262] If the flash potential is less than the minimum recharge potential (e.g. if the liquid level is less than the respective threshold for a given pressure), then the controller determines not to add subcooled water until sufficient heat has been provided to raise the flash potential to the minimum recharge potential, referred to herein as operating in the recharge mode with a flash priority.

[0263] If the flash potential is equal to or greater than the minimum recharge potential, then the controller determines to add subcooled water to increase the mass of water within the pressure vessel to the target peak mass of water. The method moves to block **820**, where in the recharge mode with water-priority, the controller determines an amount of water to supply to reach the target peak mass of water. In block **822**, where in the recharge mode with water-priority, the controller determines a rate of supplying water to the pressure vessel. In this example, the rate is selected so that the flash potential is maintained at (or above) the minimum recharge potential despite the addition of subcooled water, which is possible due to simultaneously heating water within the pressure vessel at a rate which offsets the water addition and any steam discharge. This may be achieved by calculating the rate of water addition that causes the minimum recharge potential to be maintained, for example based on a current and/or predicted power output of the heater (which itself may be limited to a power available from the power source), and a current and/or predicted demand.

[0264] In block **824**, the subcooled water is supplied to the pressure vessel at the determined amount, and the method repeats to block **802**. As above, the method may repeat before the determined amount of subcooled water is actually

provided, such that the method continually re-evaluates whether to supply subcooled liquid based on the prevailing and/or predicted conditions.

[0265] When the amount of liquid reaches the target peak mass of water and the method returns to block **808** of the method **800**, the controller determines to operate in the recharge mode with flash priority, thereby preventing further addition of subcooled liquid while allowing the storage pressure to gradually build to the target peak storage pressure.

[0266] The above description provides an example of operating the installation to phase profiles of subcooled water supply and heating to front-load water supply (while maintaining a minimum recharge flash potential). As stated elsewhere herein, this may result in the temperature of the water being maintained low for a relative long period of the recharge period, only rising towards to a temperature corresponding to the storage period in a final phase of operation. This may reduce heat loss through the walls of the pressure vessel.

[0267] Reference should again be made to the example profiles of storage pressure, liquid level, flash potential and liquid level margin as described above with reference to FIGS. **3a-4b**, which depict a depletion period corresponding to operation in the depletion mode as described above, and a recharge period corresponding to operation in the recharge mode, with respective phases of flash-priority and water-priority.

[0268] In the example methods described above, and in particular the example method **500** of FIG. **5** of controlling discharge of steam to the thermal load or thermal loads, it may be that all thermal energy demands of the respective thermal loads without reference to a priority of the loads, until the flash potential (or useful flash potential) is exhausted.

[0269] However, it may be that a criterion is evaluated, for example in block **520** of the method **500** of FIG. **5**, to determine whether to meet a thermal energy demand of a respective load in order to conserve flash potential. For example, a criterion may be evaluated corresponding to whether the flash potential is sufficient to meet a predicted demand of the plurality of loads. If the evaluation corresponds to the flash potential being sufficient to meet the predicted demand, then the controller continues to operate the installation so that steam is discharged to each of the loads at a rate to meet the respective demands. However, if the evaluation corresponds to the flash potential being insufficient to meet the predicted demand, then the controller operates the installation to preferentially discharge steam to a higher priority load or loads in preference to lower priority loads. For example, the controller may store or otherwise access priority data specifying a relative priority of the respective thermal loads. For example, a sterilization load may be of relatively higher priority (e.g. priority rating 10) than heating (e.g. priority rating 4). The priority data may be defining in any suitable way, with a simple implementation being data defining a high or low priority (e.g. with a 1 or 0 respectively).

[0270] The criterion may be any suitable criterion for assessing whether the predicted demand can be met by discharge of steam from water stored in the pressure vessel. In a relatively simple example implementation, the criterion may correspond to determining whether the flash potential is higher or lower than a predetermined amount for a respec-

tive time. For example, where an installation is commissioned to meet a relatively constant demand over a predetermined time period, such as a demand of 200 kg/hr between 09:00 and 16:00, the installation may be configured to satisfy a corresponding serviceable demand (e.g. 1400 kg) over that time period by lowering the storage pressure of the pressure vessel from peak storage conditions (i.e. with the liquid level at a target peak liquid level, and the storage pressure at a target peak storage pressure). Continuing that example, it may be expected that no more than 600 kg of steam has been discharged by 12:00, or that the storage pressure and/or liquid level in the pressure vessel at 12:00 correspond to a flash potential of at least 800 kg while maintaining a zero or positive liquid level margin.

[0271] The criterion may be defined to determine whether the storage pressure is above or below a predetermined minimum storage pressure at particular times corresponding to the serviceable demand, or is above or below a minimum storage pressure determined as a function of time. For example, there may be a minimum storage pressure of 1.8 MPa at 12:00, and a minimum storage pressure of 1.6 MPa at 14:00. The storage pressure being below the minimum storage pressure would correspond to the demand being higher than expected, and thereby indicative that the flash potential in the vessel may not be sufficient to meet a predicted demand (based on continued usage at 200 kg/hr). A similar evaluation may be performed based on any suitable parameter, for example based on storage temperature (which correlates with pressure), the liquid level (height or volume/volume fraction), or based on direct calculation/estimation of the flash potential and/or liquid level margin, for example using monitored liquid level and storage pressure information. Such an evaluation may take into account a predicted power output of the heater as described elsewhere herein.

[0272] In other example implementations, the criterion may be defined to compare a predicted demand with a serviceable demand (i.e. a demand which is known to be serviceable based on the thermal energy storage capacity of the pressure vessel). The serviceable demand may be evaluated based on current conditions of the pressure vessel (e.g. storage pressure and liquid level), or it may be a total serviceable demand corresponding to depletion from peak storage conditions (i.e. with the liquid level at a target peak liquid level, and the storage pressure at a target peak storage pressure). When the serviceable demand is expressed in terms of an amount of steam that can be discharged, it is equivalent to a flash potential when the liquid level margin is zero or positive (or equivalent to the useful flash potential as defined elsewhere herein). Similarly, the predicted demand may be a predicted demand for a future time period (e.g. starting at the moment of evaluating the criterion), or it may alternatively include a demand already met by discharge of steam during the depletion period (i.e. it may be a predicted total demand for the depletion period). In the latter case, the criterion may not refer directly to a current serviceable demand or current flash potential of the pressure vessel, but may only consider whether the predicted total demand exceeds the total serviceable demand or total flash potential corresponding to depletion from peak storage conditions.

[0273] A determination of the serviceable demand, flash potential, liquid level margin and/or useful flash potential may take into account a predicted power output of the heater

as described elsewhere herein, with the availability of heating power having the effect of slowing a drop in storage pressure to discharge a given amount of steam.

[0274] In the example installation **100** of FIG. 1, the controller **200** is configured to evaluate a criterion that is based on a predicted demand for a depletion period from 09:00-17:00 and a determined flash potential during that period, and is evaluated repeatedly through the depletion period. The predicted demand is determined as a mass of steam required during the depletion period (e.g. in kgs), with the demand corresponding to heating being a predicted demand determined by the controller as a function of forecast weather conditions, and the demand corresponding to sterilization being a predicted demand corresponding to an average of the historical demand on the same day of the week over the last 6 weeks (e.g. the 6 most recent Thursdays, if today is a Thursday). For example, the controller may receive weather forecast data including an average forecast ambient temperature for the day or over the depletion period, and may evaluate a function which factors a baseline heating demand as a function of the average forecast ambient temperature. In other examples, the heating demand and/or sterilization demand may be determined at least partly based on other data for the specific day. For example, a heating demand may be predicted based on a number of rooms which are predicted to be in use based on a central calendar system of the plant **10**, or the sterilization demand may be predicted based on a database storing information relating to sterilization jobs to be performed on the respective day.

[0275] In this example, the flash potential is determined at the respective time based on the current storage conditions, namely the current storage pressure in the pressure vessel (based on the pressure signal) and the current liquid level in the pressure vessel (based on the level signal), and is expressed as a mass of steam (e.g. in kgs). The flash potential is determined as a predicted flash potential based on the current storage conditions and based on a predicted power output of the heater as described elsewhere herein. For example, the controller **200** may receive power forecast data received by the controller from the local power source (e.g. specifying a forecast amount of power availability in a plurality of 10 minute time segments as predicted by a controller of a local power source such as a wind turbine or microgrid, based on weather forecast data), and the controller **200** may simulate a profile of storage conditions in the pressure vessel to the end of the depletion period, based on the current storage conditions, the predicted demand and the predicted power output of the heater.

[0276] In this example, the criterion corresponding to whether the flash potential is sufficient to meet the predicted demand is evaluated by determining whether the liquid level margin is predicted to go negative during the depletion period as simulated.

[0277] When the controller **200** evaluates the criterion and determines that the flash potential is insufficient to meet the predicted demand, the controller takes responsive action to reduce steam consumption by lower priority loads.

[0278] In this particular example, the controller **200** stores priority data defining a priority of each thermal load, and in particular a priority of 0 (low priority) for the first thermal load **30** (heating system) and a priority of 1 (high priority) for the second thermal load **40** (sterilization system). Based on the priority data, the controller takes responsive action by communicating a load-shedding signal to a control of the

plant **10** that controls the first thermal load, indicating a reduced capacity to meet the thermal energy demand of the load. Accordingly, the plant **10** may transition to a low power consumption mode (e.g. by setting a lower temperature set point for the system), thereby reducing the thermal energy demand.

[0279] In other examples, the controller **200** may take other responsive action, for example controlling a respective control valve for the low priority thermal load(s) to prevent discharge of steam to the respective load, irrespective of the respective thermal energy demand, or by controlling the respective control valve to discharge steam to only partially meet the respective thermal energy demand.

[0280] FIG. **9** shows a further example thermal energy storage and supply installation **900** installed for use with an industrial plant **90**. Many of the components of the installation and industrial plant are substantially as described above with reference to the example of FIG. **1**, and like reference numerals are used for like components.

[0281] In particular, like components in the industrial plant **90** include a single thermal plant load **40** which is a heating system, and which is thermally coupled to a heat exchanger **930** of the installation **900** by a closed circuit **132** as described above with reference to the example industrial plant **10** of FIG. **1**.

[0282] Like components in the installation **900** which are all as described above with reference to FIG. **1** are discussed in this paragraph and include the pressure vessel **110** provided with a heater **112** and associated power supply **113**, and containing water having a liquid level **113**. The pressure vessel is provided with a deaerator **120** fed with subcooled liquid from a subcooled water supply **150** by an inlet pump **122**. The pressure vessel **110** has an outlet **116** which discharges steam to a discharge line **101**. Sensing equipment includes a level sensor **214** and a pressure sensor **210** disposed within the pressure vessel as described above. A heat exchanger **930** is provided as a thermal load of the installation which receives steam discharged from the pressure vessel, but differs from the heat exchanger **130** as discussed below.

[0283] The installation **900** differs from that described above with reference to FIG. **1** in that the heat exchanger **930** is oriented with respect to the pressure vessel **110** to define a thermosiphon between the pressure vessel and the heat exchanger, whereby discharged steam condensing within the heat exchanger forms a column of subcooled water that is returned to a condensate inlet at a lower portion of the pressure vessel via a condensate return line **903**. In particular, the heat exchanger **930** is disposed at a higher position than the pressure vessel (and therefore higher than a peak liquid water level in the pressure vessel expressed as a height), for example by a height difference of at least 0.5 m measured between the uppermost point of a heat exchange element of the heat exchanger and the uppermost point of the pressure vessel (for example between 0.5 m and 2 m, for example 0.6 m or 1 m higher). The heat exchanger may be disposed at a higher position than the pressure vessel by virtue of a condensation location in the heat exchanger being at a higher position than the uppermost point of the pressure vessel or than an operational range of fill levels in the pressure vessel, for example by a height difference of at least 0.1 m, for example at least 0.2 m, for example between 0.1 m and 2 m. The controller **920** is configured to discharge steam to the heat exchanger with a minimal pressure drop

from the storage pressure such that, given the relative position of the heat exchanger, there is adequate head to return the subcooled water to the pressure vessel at any fill level under the action of gravity. It is thought that the action of the heat exchanger to condense the steam promotes a continuing flow of steam from the pressure vessel in the manner of a thermosiphon. Further, subcooling in the heat exchanger **930** increases the density of the water and promotes recirculation back to the pressure vessel **110**.

[0284] The controller may control the control valve **915** to open fully so that steam is discharged substantially at the storage pressure. In the event of excessive steam flow for the thermal load, the controller may control the control valve to open at a duty cycle to meter the flow over a time period (e.g. open 60% of the time and closed 40% of the time in a given time period). In a variant embodiment, instead of there being a discharge control valve **915** at the outlet of the pressure vessel, or in addition to that control valve **915**, there may be a condensate line control valve **915'** disposed along the condensate return line **903** which conveys the condensate from the heat exchanger **930** to the pressure vessel **110**. The controller may be operatively coupled (not shown) to control the condensate line control valve **915'** to meter (e.g. selectively permit) the column of condensate formed in and downstream of the heat exchanger **930** to return to the pressure vessel. When the condensate line control valve **915'** is closed, condensate accumulates in the heat exchanger **930** such that no further steam can enter the heat exchanger **930**, thereby limiting discharge of steam from the pressure vessel.

[0285] Any pressure reduction by the control valve **915** at the outlet of the pressure vessel (where present) can be controlled to vary the pressure exerted by the column to permit condensate to return to the pressure vessel. Similarly, an amount of subcool at the heat exchanger **930** can be controlled to vary density of the subcooled water to control the thermosiphon flow. Further still, a suitable height can be selected for any particular installation by modelling to ensure continuing operation as a thermosiphon over a desired range of conditions.

[0286] As there is only a single thermal load **930**, in this example there is a single discharge control valve **915** operatively coupled to the controller along the discharge line **101**, for controlling discharge of steam to the heat exchanger **930** (which nevertheless is optional as explained above). Downstream of the outlet/discharge control valve **915** there is a flowmeter **916** and a pressure sensor **918** coupled to a controller **920** to provide a flow signal and a pressure signal respectively. In this simplified example, the discharge line **101** discharges steam only to a single distribution line **902** that extends to the heat exchanger **930**, but which is separately described as it will be appreciated that in variant examples there may be a plurality of branching distribution lines.

[0287] The controller **920** is substantially as described above with reference to the controller **200** of FIG. **1** in respect of determining a demand based on a demand signal from the thermal plant load **40** and in controlling heating of water within the pressure vessel using the heater **112**. The controller **920** is configured to control discharge of steam based on the demand signal received from the thermal plant load **40** by controlling the discharge control valve **915**. In this example, the controller **920** causes the discharge control valve **915** to open to discharge steam at a variable discharge pressure which is at or marginally below the storage pres-

sure, to provide a flow rate of steam that permits a target rate of heat transfer within the heat exchanger **930** that satisfies the demand. As with the heat exchanger **130** of FIG. 1, the controller **920** is configured to activate a heat transfer pump **133** associated with the heat transfer circuit **132** upon selectively discharging steam to the heat exchanger **930** for heat transfer. In the variant example mentioned above, the controller **920** may control a condensate line control valve **915'** to control discharge of steam to the heat exchanger, at a flow rate of stem that permits a target rate of heat transfer within the heat exchanger **930** that satisfies the demand.

[0288] As there is a thermosiphon arrangement for the return of condensate to the pressure vessel, subcooled water is returned to the pressure vessel naturally, without necessitating control by the controller **920** unless the controller limits the return flow in order to control discharge of steam from the pressure vessel. The controller **920** is configured to control an initial supply of subcooled water to the pressure vessel from a water supply via the deaerator **120** as described above, which in this example is shown as a water supply vessel **150** but in other examples may be another water source.

[0289] In use, an initial supply of water is provided to the pressure vessel **110** from the water source to a target peak water mass as described above. The controller **920** operates the heater **112** to heat the water to the target peak storage pressure, such as at least 2 MPa. In response to a demand for thermal energy (e.g. based on the demand signal received from the thermal plant load **40**), the controller **920** controls the discharge control valve **915** to open to lower the storage pressure within the pressure vessel and thereby flash liquid water within the pressure vessel, which is selectively discharged via the discharge control valve **915**. Alternatively, a condensate line control valve **915'** can be controlled to permit steam to be discharged from the vessel.

[0290] The steam is provided to the heat exchanger **930** located above the pressure vessel **110** at a discharge pressure, condensing within the heat exchanger **930** to give up its latent heat of vaporisation for heat transfer to a working fluid in the heat transfer circuit **132**, which then transfers heat to the thermal plant load **40** (a heating system). In response to the demand for thermal energy, the controller **920** also operates the pump **133** of the heat transfer circuit **132** ready for heat transfer from steam to be received at the heat exchanger **930**.

[0291] The steam is subcooled within the heat exchanger **930**, for example by at least 10° C., and returns to the pressure vessel **110** by a thermosiphon mechanism owing to a pressure head of a column of subcooled water between the heat exchanger **930** and the pressure vessel (together with any velocity head owing to a flow rate of the steam). Condensate returning to the pressure vessel may cause thermal stratification of liquid water within the pressure vessel, such that liquid water at the interface with steam remains at the saturate temperature for immediate flashing upon lowering of the storage pressure by opening the respective discharge control valve.

[0292] The controller **920** independently controls the heater **112** to heat water within the pressure vessel by the method **600** of FIG. 6.

[0293] As described above, the installation **900** can be operated to discharge and subsequently recharge a large reserve of thermal energy, for example in a depletion period and a recharge period as described above. In particular,

steam may be discharged to the thermal load **930** during a depletion period at a rate such that an enthalpy loss from the pressure vessel (taking into account an enthalpy of a returning flow of subcooled water from the heat exchanger **930**) is greater than an input energy provided to water within the pressure vessel via the heater, and this may result in the storage pressure of water within the pressure vessel reducing from a target peak storage pressure (e.g. of at least 2 MPa) reducing significantly, for example by at least 1 MPa. Similarly, during a period of no or relatively low demand, an input energy from the heater may be greater than any enthalpy loss owing to discharge of steam (again taking into account an enthalpy of a returning flow of subcooled water from the heat exchanger **930**), such that the storage pressure rises over a recharge period, and may rise by a significant amount for example at least 1 MPa.

[0294] As will be appreciated, operation of the thermosiphon installation **900** of FIG. 9 does not involve any procedures for determining a phasing of subcooled water supply, since subcooled water is continually returned from the heat exchanger **930** via the thermosiphon arrangement,

[0295] A thermal energy storage and supply installation as described herein may have a connection to a non-local electrical grid (referred to herein as “grid power”, whereas power drawn from a microgrid is considered to be drawn from a local power source), and a controller of the installation may be configured to evaluate a criterion for drawing grid power based on a criterion corresponding to either (i) a predicted power output of the heater based on power supply from the local power source being insufficient to meet a thermal energy demand during a depletion period; and/or (ii) a predicted power output of the heater based on power supply from the local power source being insufficient to meet a target peak storage conditions (e.g. a target peak storage pressure for a target peak mass of water) during a recharge period.

[0296] When a demand for supplemental grid power is determined based on (i) or (ii) the controller may determine when and/or whether to draw grid power based on cost. For example, it may be that power cost is relatively high during the day and relatively low during the night. Accordingly, the criterion may be biased to permit power draw from the grid power source when the power cost is relatively low. The criterion may be based on current or predicted conditions of the pressure vessel, for example a current or predicted flash potential or liquid level margin over a time period. As noted elsewhere herein, a predicted flash potential may be based on a predicted demand for thermal energy and/or a predicted power output of the heater (corresponding to a predicted availability of power for heating). Accordingly, the criterion may permit power draw from the power source at relatively higher costs when it is determined that the ability of the installation to meet a demand would otherwise be compromised. Additionally or alternatively, the criterion may be defined so that when there is a demand for supplemental grid power but energy costs are relatively high, steam supply to relatively lower priority thermal loads is restricted while being maintained for higher priority thermal loads, as described elsewhere herein.

[0297] By way of example only, the installation may have a local power source comprising a wind power source and a grid connection to a non-local electrical grid, and the associated industrial plant may be operated such that there is a relatively high demand for thermal energy during daytime

hours each day starting from 09:00, with minimal thermal energy demand between 18:00 and 09:00. Based on conditions within the pressure vessel at 18:00, the controller may determine that 700 kW of heating is to be supplied in an overnight recharge period in order to reach the target peak storage pressure and target liquid water level by 09:00. At the start of the recharge period, the controller may predict a power availability from the wind power source for the heater of 900 kW, based on historical power availability data for the heater and/or based on forecast weather conditions (both of which may be determined by reference to historical power availability data and/or weather forecast data provided to the controller or stored in a database accessible by the controller). At an intermediate time within the recharge period (such as 03:00), the controller may determine that the predicted power availability over the recharge period is only 600 kW, for example based on updated weather forecast conditions. Based on this determination, the controller may determine to draw power supplementary power from the grid power source. The controller may determine a latest time at which the supplementary power draw may commence, based on current and/or predicted conditions in the pressure vessel, and delay drawing the supplementary power until that latest time. In particular, the heater may have a maximum power output (e.g. 120 kW). Further, it may be that a rate of introducing subcooled liquid to the pressure vessel is limited by an amount of concurrent heating (e.g. to maintain a flash potential while permitting a liquid level margin to rise), and/or by a rate at which such subcooled liquid can introduced while adequately deaerated.

[0298] The disclosure extends to the subject-matter of the following numbered examples:

[0299] Example 1. A method of thermal energy storage and supply, comprising:

[0300] providing subcooled water to a pressure vessel;

[0301] heating liquid water within the pressure vessel using an electrically-powered heater so that the vessel contains saturated liquid water and steam at a variable storage pressure;

[0302] controlling the heater to raise the storage pressure to a peak storage pressure, wherein the peak storage pressure is at least 0.5 MPa, for example at least 1 MPa or at least 2 MPa;

[0303] selectively discharging steam from an outlet of the pressure vessel to a thermal load, in response to a thermal energy demand, such that during a depletion period the storage pressure reduces from the peak storage pressure, for example by at least 50% of the peak storage pressure.

[0304] Example 2. A method according to Example 1, wherein:

[0305] the discharged steam is provided to the thermal load directly without passing through an intermediate steam accumulator; or

[0306] any steam accumulator or steam accumulators between the pressure vessel and the thermal load have a total volume which is less than a volume of the pressure vessel.

[0307] Example 3. A method according to Example 1 or 2, comprising operating the heater during the depletion period, optionally simultaneously with selectively discharging steam.

[0308] Example 4. A method according to any preceding Example, comprising operating the heater without simultaneously providing subcooled water to the pressure vessel;

[0309] for part or all of the depletion period; and/or

[0310] for part or all of a recharge period, in which the storage pressure increases to the peak storage pressure, for example by at least 50% of the peak storage pressure.

[0311] Example 5. A method according to any preceding Example, wherein steam is discharged from the outlet so that a liquid level of liquid water within the pressure vessel falls below a lower limit liquid level for operation of the heater during the depletion period;

[0312] optionally wherein the controller operates the heater during the depletion period simultaneously with steam being discharged when the liquid level of water is above the lower limit liquid level, and wherein the controller deactivates the heater for continued discharge of steam beyond the lower limit liquid level.

[0313] Example 6. A method according to any preceding Example, wherein steam is discharged to the thermal load at a discharge pressure; and

[0314] wherein subcooled water is provided to the vessel when the storage pressure is greater than the discharge pressure.

[0315] Example 7. A method according to Example 6, wherein the peak storage pressure exceeds a maximum water supply pressure at which subcooled water is provided to the vessel.

[0316] Example 8. A method according to Example 6 or 7, wherein the subcooled water is provided to the pressure vessel during the depletion period; optionally wherein a mass of subcooled water provided to the pressure vessel during the depletion period is no more than 50% of a mass of steam discharged during the depletion period, for example no more than 25%, no more than 10% or no more than 5%.

[0317] Example 9. A method according to any of Examples 1-6, wherein the thermal load comprises a heat exchanger oriented with respect to the pressure vessel so that a thermosiphon is established between the pressure vessel and the heat exchanger, such that discharged steam condensing within the heat exchanger forms a column of subcooled water that is returned to a condensate inlet at a lower portion of the pressure vessel, during the depletion period.

[0318] Example 10. A method according to Example 9, wherein:

[0319] the heat exchanger is disposed at a higher position than the condensate inlet to give adequate head to return the subcooled water to the pressure vessel under the action of gravity; and/or

[0320] wherein the heat exchanger subcools the water within the heat exchanger by at least 10° C., relative to a saturation temperature corresponding to the storage pressure

[0321] Example 11. A method according to any of Examples 6 to 10, wherein the subcooled water is provided to the pressure vessel at an inlet flow rate such that the storage pressure reduces as the subcooled water is provided.

[0322] Example 12. A method according to any preceding Example, wherein a flash potential corresponds to an amount of steam that can be flashed from liquid water within the pressure vessel before the storage pressure reaches a lower limit pressure for sustaining discharge of steam to the thermal load; and

[0323] wherein a liquid level margin corresponds to an amount of liquid water within the pressure vessel above a lower limit amount for operation of the heater upon discharge of the flash potential;

[0324] the method comprising:

[0325] evaluating a criterion corresponding to whether the liquid level margin is positive or negative;

[0326] based on the evaluation, providing subcooled water to the pressure vessel to increase the liquid level margin and to reduce the flash potential.

[0327] Example 13. A method according to Example 12, wherein the flash potential is a predicted flash potential which is a function of a predicted demand and/or a predicted power output of the heater over a time period, and wherein the liquid level margin is a predicted liquid level margin based on discharge of the predicted flash potential over the time period.

[0328] Example 14. A method according to Example 13, wherein the predicted demand is based on historical demand data, such as historical steam or thermal demand data for an apparatus (e.g. an installation); and/or

[0329] wherein the predicted demand is based on forecast weather conditions; and/or

[0330] wherein the predicted power output of the heater is based on historical power availability data for the heater; and/or

[0331] wherein the predicted power output of the heater is based on forecast weather conditions.

[0332] Example 15. A method according to any preceding Example, wherein there is a plurality of thermal loads including a first thermal load and a second thermal load;

[0333] wherein steam is selectively discharged from the pressure vessel to each of the thermal loads via respective control valves, based on respective thermal energy demands;

[0334] wherein a flash potential corresponds to an amount of steam that can be flashed from liquid water within the pressure vessel before the storage pressure reaches a lower limit pressure for sustaining discharge of steam to the thermal load; the method further comprising:

[0335] evaluating a criterion corresponding to whether the flash potential is sufficient to meet a predicted demand of the plurality of loads;

[0336] based on the evaluation and on priority data relating to the thermal loads, determining to discharge steam to the first thermal load to meet a respective first thermal energy demand in preference to discharging steam to the second thermal load to meet a respective second thermal energy demand.

[0337] Example 16. A method according to any preceding Example comprising, during a recharge period:

[0338] heating liquid in the pressure vessel to raise the storage pressure to the peak storage pressure by a recharge pressure difference which is at least 50% of the peak storage pressure;

[0339] providing subcooled water to the pressure vessel to reach a peak mass of water in the pressure vessel corresponding to a peak liquid level at the peak storage pressure;

[0340] optionally wherein a duration of the recharge period is at least as 100% of the duration of the depletion period, for example at least 125% or at least 150%.

[0341] Example 17. A method according to Example 16, comprising phasing a profile of water supply during the recharge period so that it is front-loaded relative to a profile of heating during the recharge period.

[0342] Example 18. A method according to Example 16 or 17, wherein a flash potential corresponds to an amount of steam that can be flashed from liquid water within the pressure vessel before the storage pressure reaches a lower limit pressure for sustaining discharge of steam to the thermal load; and

[0343] wherein a liquid level margin corresponds to an amount of liquid water within the pressure vessel above a lower limit amount for operation of the heater upon discharge of the flash potential;

[0344] wherein the method comprises, during a water-priority portion of the recharge period, phasing a profile of water supply relative to a profile of heating to maintain a minimum recharge flash potential while increasing an amount of water within the pressure vessel to the target peak mass of water, such that the liquid level margin progressively rises; and

[0345] subsequently heating the liquid water during a flash-priority portion of the recharge period to raise the storage pressure to the peak storage pressure.

[0346] Example 19. A method according to Example 18, wherein when the minimum recharge flash potential corresponds to a predicted demand, and the predicted demand corresponds to no demand for steam during an idle portion of the recharge period, the method comprises providing subcooled water to reduce the storage pressure below the lower limit pressure during the idle portion, the pressure subsequently rising owing to heat input to provide a flash potential corresponding to a non-zero demand.

[0347] Example 20. A method according to any preceding Example, wherein a maximum flow rate of steam discharge during the depletion period per unit area of a liquid level of water in the pressure vessel is no more than $150 \text{ kg/m}^2\text{h}$, for example no more than $100 \text{ kg/m}^2\text{h}$ or no more than $50 \text{ kg/m}^2\text{h}$.

[0348] Example 21. A method according to any preceding Example, comprising deaerating an inlet flow of subcooled water provided to the pressure vessel along a deaeration path by conveying a deaerating flow of steam from within the pressure vessel past in counterflow along the deaeration path;

[0349] wherein the inlet flow is provided into the pressure vessel over a range of storage pressures of the pressure vessel;

[0350] wherein a velocity of the deaerating flow of steam is varied as a function of a temperature of the inlet flow and/or a temperature difference between the inlet flow and the steam to target the inlet flow reaching a saturation temperature corresponding to the steam over the range of storage pressures;

[0351] optionally wherein the velocity is varied by controlling a control valve for venting the deaerating flow of steam and associated entrained gases;

[0352] optionally wherein the velocity or a control valve setting for controlling the velocity is determined by reference to a database of velocities or control valve settings correlated to the temperature of the inlet flow and/or the temperature of the steam.

[0353] Example 22. A method according to any preceding Example, wherein:

[0354] the peak storage pressure is at least 2.5 MPa, for example at least 3 MPa; and/or

[0355] wherein the storage pressure reduces during the depletion period to a value of no more than 1.5 MPa, for example no more than 1 MPa, or no more than 0.8 MPa.

[0356] Example 23. A method according to any preceding Example, wherein:

[0357] an average depletion power is defined as a cumulative enthalpy of steam discharged from the pressure vessel during the depletion period divided by a duration of the depletion period; and

[0358] a maximum depletion power is defined as a maximum enthalpy of steam discharged within the depletion period during any minimum power evaluation period of one minute, divided by the minimum power evaluation period.

[0359] Example 24. A method according to Example 23, further comprising:

[0360] heating liquid water within the pressure vessel, using the heater, during a recharge period in which the storage pressure increases by at least 50% of the peak stage pressure;

[0361] wherein an average recharge power is defined as a cumulative energy provided to the liquid water by the heater during the recharge period, divided by a duration of the recharge period;

[0362] wherein a maximum reheat power is defined as a maximum power at which energy is provided to the liquid water by the heater within the recharge period;

[0363] wherein the average reheat power is no more than 50% of the average depletion power; and/or

[0364] wherein the average reheat power is no more than 50% of the maximum depletion power; and/or

[0365] wherein the maximum reheat power is no more than 50% of the maximum depletion power.

[0366] Example 25. A method according to Example 23 or 24, wherein a dimensional ratio of the cumulative enthalpy of (i) steam discharged from the pressure vessel during the depletion period and (ii) the average reheat power of the heater during the recharge period is at least 25000 seconds.

[0367] Example 26. A method according to any preceding Example conducted using a thermal energy storage and supply apparatus (e.g. an installation) in accordance with any of Examples 27 to 47.

[0368] Example 27. A thermal energy storage and supply apparatus (e.g. installation) comprising:

[0369] a pressure vessel for storing water comprising saturated liquid water and steam at a storage pressure of 0.5 MPa, for example 1 MPa or 2 MPa, the pressure vessel having an outlet to discharge steam to a thermal load;

[0370] an electrically-powered heater configured to heat liquid water stored in the pressure vessel to vary a storage pressure within the pressure vessel;

[0371] a controller configured to operate the thermal storage installation by:

[0372] controlling the heater to heat liquid water within the pressure vessel to reach a peak storage pressure of saturated liquid water and steam of at least 0.5 MPa, for example at least 1 MPa or at least 2 MPa;

[0373] controlling a control valve to selectively discharge steam from the outlet to a thermal load, in response to a thermal energy demand;

[0374] permitting steam discharge to meet the thermal energy demand so that the storage pressure reduces from the peak storage pressure by a depletion pressure difference of at least 50% of the peak storage pressure.

[0375] Example 28. An apparatus according to Example 27, wherein the outlet is in communication with the thermal load to provide the discharged steam directly to the thermal load without passing through an intermediate steam accumulator; or

[0376] wherein any steam accumulator or steam accumulators between the pressure vessel and the thermal load have a total volume which is less than a volume of the pressure vessel.

[0377] Example 29. An apparatus according to Example 27 or 28, wherein the controller is configured to control the heater to heat liquid water within the pressure vessel independently of controlling the control valve to selectively discharge steam, whereby in use simultaneous heating and discharge of steam is permitted, and each of heating and discharge of steam is permitted without the other.

[0378] Example 30. An apparatus according to any of Example 27 to 29, wherein the controller is configured to control the heater to heat liquid water within the pressure vessel independently of causing subcooled water to be supplied to the pressure vessel, whereby in use simultaneous heating and subcooled water supply is permitted, and heating without simultaneous subcooled water supply is permitted.

[0379] Example 31. An apparatus according to any of Example 27 to 30, wherein the controller is configured to cause subcooled water to be supplied to the pressure vessel when the storage pressure is greater than a discharge pressure to which steam is discharged from the outlet.

[0380] Example 32. An apparatus according to any of Examples 27 to 31, further comprising a water pump configured to supply subcooled water to the pressure vessel; optionally wherein the water pump has a maximum water supply pressure which is lower than the peak storage pressure.

[0381] Example 33. An apparatus according to any of Examples 27 to 32, further comprising a subcooled water supply vessel configured to store subcooled water for supply to the pressure vessel; wherein a ratio of a storage volume of the subcooled water supply vessel to a storage volume of the pressure vessel is at least 5%; optionally at least 7.5% or at least 10%.

[0382] Example 34. An apparatus according to any of Examples 26 to 33, wherein the controller is configured to selectively operate in an extended depletion mode in which the controller permits steam to be discharged so that a liquid level of liquid water within the pressure vessel falls below a lower limit liquid level for operation of the heater during the depletion period, wherein the controller prevents heating with the heater when the liquid level is below the lower limit liquid level for operation of the heater;

[0383] optionally wherein the controller is configured to selectively operate the heater in a heated depletion mode in which the controller prevents steam discharge that would cause the liquid level to fall below the lower

limit liquid level; wherein the controller permits heating with the heater when the liquid level is below the lower limit liquid level.

[0384] Example 35. An apparatus according to any of Examples 27 to 31, wherein the thermal load comprises a heat exchanger oriented with respect to the pressure vessel to define a thermosiphon between the pressure vessel and the heat exchanger, whereby discharged steam condensing within the heat exchanger forms a column of subcooled water that is returned to a condensate inlet at a lower portion of the pressure vessel.

[0385] Example 36. An apparatus according to Example 35, wherein the heat exchanger is disposed at a higher position than the pressure vessel and the controller is configured to cause steam to be discharged to the heat exchanger at a discharge pressure selected such that, given the relative position of the heat exchanger, there is adequate head to return the subcooled water to the pressure vessel under the action of gravity; and/or

[0386] wherein the controller is configured to control heat exchange at the heat exchanger such that the water is subcooled by at least 10° C.

[0387] Example 37. An apparatus according to any of Examples 27 to 36, comprising a level sensor configured to provide a level signal corresponding to a liquid level of water within the pressure vessel to the controller; and/or

[0388] comprising a storage pressure sensor configured to provide a pressure signal corresponding to a pressure of water within the pressure vessel to the controller; and/or

[0389] comprising a storage temperature signal configured to provide a temperature signal corresponding to a temperature of water within the pressure vessel to the controller; and/or

[0390] comprising a discharge pressure sensor configured to provide a pressure signal to the controller corresponding to a discharge pressure to which steam discharged from the outlet of the pressure vessel; and/or

[0391] comprising a discharge flowmeter configured to provide a flow rate signal to the controller corresponding to a flow rate of steam downstream of the outlet; and/or

[0392] comprising an inlet flowmeter configured to provide a flow rate signal to the controller corresponding to a flow rate of subcooled water provided to the pressure vessel.

[0393] Example 38. An apparatus according to any of Examples 27 to 37, wherein a flash potential corresponds to an amount of steam that can be flashed from liquid water within the pressure vessel before the storage pressure reaches a lower limit pressure for sustaining discharge of steam to the thermal load; and

[0394] wherein a liquid level margin corresponds to an amount of liquid water within the pressure vessel above a lower limit amount for operation of the heater upon discharge of the flash potential;

[0395] wherein the controller is configured to select between operating the thermal storage apparatus in a recharge mode or a depletion mode based on a predetermined setting such as a time-dependent setting, based on a user input, or based on a predicted demand profile and/or a predicted power output profile of the heater;

[0396] wherein in the depletion mode, the controller is configured to evaluate a criterion corresponding to whether the liquid level margin is positive or negative; and to provide subcooled water to the pressure vessel to increase the liquid level margin and reduce the flash potential when the evaluation corresponds to the liquid level margin being negative; and/or

[0397] wherein in the recharge mode, the controller is configured to phase a profile of water supply relative to a profile of heating to (i) maintain a minimum recharge flash potential which is a predetermined flash potential or which corresponds to a predicted demand, while a liquid level margin and an amount of water in the pressure vessel progressively increases to a target mass of water corresponding to a target liquid level at the peak storage pressure; and (ii) subsequently heat the liquid water to raise the storage pressure to the peak storage pressure.

[0398] Example 39. An apparatus according to Example 38, wherein the flash potential is a predicted flash potential, and wherein the controller is configured to predict the flash potential as a function of a predicted demand and/or a predicted power output of the heater over a time period;

[0399] wherein the liquid level margin is a predicted liquid level margin, wherein the controller is configured to predict the liquid level margin based on discharge of the predicted flash potential over the time period.

[0400] Example 40. An apparatus according to Example 38 or 39, wherein the controller is configured to determine the predicted demand based on historical demand data, such as historical steam or thermal demand data for the apparatus; and/or

[0401] wherein the controller is configured to determine the predicted demand based on forecast weather conditions; and/or

[0402] wherein the controller is configured to determine the predicted power output of the heater based on historical power availability data for the heater; and/or

[0403] wherein the controller is configured to determine the predicted power output of the heater based on forecast weather conditions.

[0404] Example 41. An apparatus according to any of Examples 38-40, wherein in the recharge mode, the controller is configured so that when the predicted demand corresponds to no demand for steam during an idle portion of a recharge period, the controller causes subcooled water to be supplied to reduce the storage pressure below the lower limit pressure during the idle portion;

[0405] optionally wherein the controller is configured to subsequently raise the storage pressure by heating liquid water within the pressure vessel to provide a flash potential corresponding to a non-zero demand.

[0406] Example 42. An apparatus according to any of Examples 27 to 41, wherein the controller is configured to control the control valve so that steam is selectively discharged at a maximum flow rate per unit area of a liquid level of water in the pressure vessel which is no more than 150 kg/m²h, for example no more than 100 kg/m²h or no more than 50 kg/m²h.

[0407] Example 43. An apparatus according to any of Examples 27 to 41, wherein the pressure vessel is configured

to discharge steam to a plurality of thermal loads via respective control valves, including a first thermal load and a second thermal load;

[0408] wherein the controller is configured to evaluate a criterion corresponding to whether the flash potential is sufficient to meet a predicted demand of the plurality of loads;

[0409] wherein, responsive to an outcome of the evaluation, the controller is configured to evaluate priority data specifying relative priorities of the thermal loads, and to operate in a priority discharge mode in which it controls selective discharge of steam to meet a thermal energy demand of a relatively high priority thermal load in preference to controlling selective discharge of steam to meet a thermal energy demand of a relatively lower priority thermal load.

[0410] Example 44. An apparatus according to any of Examples 27 to 43, configured to provide water in the form of steam discharged from the pressure vessel to the thermal load in an open arrangement, by which the thermal load consumes the water provided as steam without corresponding return of the water as condensate; and

[0411] wherein the controller is configured to discharge steam to the thermal load at a minimum discharge pressure of between 0.5 MPa and 1.0 MPa, or the thermal load is configured to receive discharged steam at a minimum discharge pressure of between 0.5 MPa and 1.0 MPa.

[0412] Example 45. An apparatus according to any of Examples 27 to 44, configured to supply steam discharged from the pressure vessel to at least two thermal loads;

[0413] optionally wherein:

[0414] the thermal storage apparatus is configured to provide water in the form of steam discharged from the pressure vessel to one of the thermal loads in an open arrangement, by which the water is placed into contact with a foreign process fluid or article, and/or is discharged from the thermal load without corresponding return of the water as condensate; and/or

[0415] the thermal storage apparatus is configured to provide water in the form of steam discharged from the pressure vessel to one of the thermal loads in a closed loop, by which the water is at least partly returned to the pressure vessel as subcooled water, for example via a subcooled supply vessel.

[0416] Example 46. An apparatus according to any of Examples 27 to 45, wherein the pressure vessel is provided with a deaerator configured to receive an inlet flow of subcooled water along a deaeration path, and configured to direct a deaerating flow of steam from within the pressure vessel in counterflow along the deaeration path;

[0417] wherein the controller is configured to vary a velocity of the deaerating flow of steam as a function of a temperature of the inlet flow and/or a temperature of the steam, to target the inlet flow reaching a saturation temperature corresponding to the steam along the deaeration path, over a range of storage pressures;

[0418] optionally wherein the controller is configured to vary the velocity by controlling a control valve for venting the deaerating flow of steam and associated entrained gases from the deaerator; and

[0419] optionally wherein the controller is configured to control the velocity or a control valve setting for controlling the velocity by reference to a database of

velocities or control valve settings correlated to the temperature of the inlet flow and/or the temperature of the steam.

[0420] Example 47. An apparatus according to any of Examples 27 to 46, wherein the heater is installed within the pressure vessel so that a lower limit liquid level for operation of the heater corresponds to a liquid fraction of water within the pressure vessel of no more than 60%, for example no more than 50%, no more than 40% or no more than 30%.

1. A method of thermal energy storage and supply, comprising:

providing subcooled water to a pressure vessel;

heating liquid water within the pressure vessel using an electrically-powered heater so that the vessel contains saturated liquid water and steam at a variable storage pressure; controlling the heater to raise the storage pressure to a peak storage pressure of at least 2 MPa; selectively discharging steam from an outlet of the pressure vessel to a thermal load, in response to a thermal energy demand, such that during a depletion period the storage pressure reduces by at least 1 MPa from the peak storage pressure.

2. A method according to claim 1, wherein:

the discharged steam is provided to the thermal load directly without passing through an intermediate steam accumulator; or

any steam accumulator or steam accumulators between the pressure vessel and the thermal load have a total volume which is less than a volume of the pressure vessel.

3-5. (canceled)

6. A method according to claim 1, wherein steam is discharged to the thermal load at a discharge pressure;

wherein subcooled water is provided to the vessel when the storage pressure is greater than the discharge pressure; and

wherein the subcooled water is provided to the pressure vessel during the depletion period.

7-14. (canceled)

15. A method according to claim 1, wherein there is a plurality of thermal loads including a first thermal load and a second thermal load;

wherein steam is selectively discharged from the pressure vessel to each of the thermal loads via respective control valves, based on respective thermal energy demands;

wherein a flash potential corresponds to an amount of steam that can be flashed from liquid water within the pressure vessel before the storage pressure reaches a lower limit pressure for sustaining discharge of steam to the thermal load; the method further comprising:

evaluating a criterion corresponding to whether the flash potential is sufficient to meet a predicted demand of the plurality of loads;

based on the evaluation and on priority data relating to the thermal loads, determining to discharge steam to the first thermal load to meet a respective first thermal energy demand in preference to discharging steam to the second thermal load to meet a respective second thermal energy demand.

16. A method according to claim 1 comprising, during a recharge period:

heating liquid in the pressure vessel to raise the storage pressure by at least 1 MPa to a peak storage pressure of at least 2 MPa;

providing subcooled water to the pressure vessel to reach a peak mass of water in the pressure vessel corresponding to a peak liquid level at the peak storage pressure; and

phasing a profile of water supply during the recharge period so that it is front-loaded relative to a profile of heating during the recharge period.

17. (canceled)

18. A method according to claim 16, wherein a flash potential corresponds to an amount of steam that can be flashed from liquid water within the pressure vessel before the storage pressure reaches a lower limit pressure for sustaining discharge of steam to the thermal load; and

wherein a liquid level margin corresponds to an amount of liquid water within the pressure vessel above a lower limit amount for operation of the heater upon discharge of the flash potential;

wherein the method comprises, during a water-priority portion of the recharge period, phasing a profile of water supply relative to a profile of heating to maintain a minimum recharge flash potential while increasing an amount of water within the pressure vessel to the target peak mass of water, such that the liquid level margin progressively rises; and

subsequently heating the liquid water during a flash-priority portion of the recharge period to raise the storage pressure to the peak storage pressure.

19-24. (canceled)

25. A method according to claim 1, comprising a recharge period in which liquid in the pressure vessel is heated to raise the storage pressure by at least 1 MPa to a peak storage pressure of at least 2 MPa;

wherein a dimensional ratio of (i) the cumulative enthalpy of (i) steam discharged from the pressure vessel during the depletion period and (ii) an average reheat power of the heater during the recharge period is at least 25000 seconds; wherein the average reheat power is defined as a cumulative energy provided to the liquid water during the recharge period, divided by a duration of the recharge period.

26. (canceled)

27. A thermal energy storage and supply installation comprising:

a pressure vessel for storing water comprising saturated liquid water and steam at a storage pressure of 2 MPa, the pressure vessel having an outlet to discharge steam to a thermal load;

an electrically-powered heater configured to heat liquid water stored in the pressure vessel to vary a storage pressure within the pressure vessel;

a controller configured to operate the thermal storage installation by:

controlling the heater to heat liquid water within the pressure vessel to reach a peak storage pressure of saturated liquid water and steam of at least 2 MPa;

controlling a control valve to selectively discharge steam from the outlet to a thermal load, in response to a thermal energy demand;

permitting steam discharge to meet the thermal energy demand so that the storage pressure reduces by at least 1 MPa from the peak storage pressure.

28-32. (canceled)

33. An installation according to claim 27, further comprising a subcooled water supply vessel configured to store subcooled water for supply to the pressure vessel;

wherein a ratio of a storage volume of the subcooled water supply vessel to a storage volume of the pressure vessel is at least 5%.

34. An installation according to claim 27, wherein the controller is configured to selectively operate in an extended depletion mode in which the controller permits steam to be discharged so that a liquid level of liquid water within the pressure vessel falls below a lower limit liquid level for operation of the heater during the depletion period, wherein the controller prevents heating with the heater when the liquid level is below the lower limit liquid level for operation of the heater.

35. An installation according to claim 27, wherein the thermal load comprises a heat exchanger oriented with respect to the pressure vessel to define a thermosiphon between the pressure vessel and the heat exchanger, whereby discharged steam condensing within the heat exchanger forms a column of subcooled water that is returned to a condensate inlet at a lower portion of the pressure vessel.

36. An installation according to claim 35, wherein the heat exchanger is disposed at a higher position than the pressure vessel and the controller is configured to cause steam to be discharged to the heat exchanger at a discharge pressure selected such that, given the relative position of the heat exchanger, there is adequate head to return the subcooled water to the pressure vessel under the action of gravity; and/or

wherein the controller is configured to control heat exchange at the heat exchanger such that the water is subcooled by at least 10° C.

37. (canceled)

38. An installation according to claim 27, wherein a flash potential corresponds to an amount of steam that can be flashed from liquid water within the pressure vessel before the storage pressure reaches a lower limit pressure for sustaining discharge of steam to the thermal load; and

wherein a liquid level margin corresponds to an amount of liquid water within the pressure vessel above a lower limit amount for operation of the heater upon discharge of the flash potential;

wherein the controller is configured to select between operating the thermal storage installation in a recharge mode or a depletion mode based on a predetermined setting such as a time-dependent setting, based on a user input, or based on a predicted demand profile and/or a predicted power output profile of the heater;

wherein in the depletion mode, the controller is configured to evaluate a criterion corresponding to whether the liquid level margin is positive or negative; and to provide subcooled water to the pressure vessel to increase the liquid level margin and reduce the flash potential when the evaluation corresponds to the liquid level margin being negative; and/or

wherein in the recharge mode, the controller is configured to phase a profile of water supply relative to a profile of heating to (i) maintain a minimum recharge flash potential which is a predetermined flash potential or which corresponds to a predicted demand, while a liquid level margin and an amount of water in the pressure vessel

progressively increases to a target mass of water corresponding to a target liquid level at the peak storage pressure; and (ii) subsequently heat the liquid water to raise the storage pressure to the peak storage pressure.

39-40. (canceled)

41. An installation according to claim 38, wherein in the recharge mode, the controller is configured so that when the predicted demand corresponds to no demand for steam during an idle portion of a recharge period, the controller causes subcooled water to be supplied to reduce the storage pressure below the lower limit pressure during the idle portion.

42. (canceled)

43. An installation according to claim 27, wherein the pressure vessel is configured to discharge steam to a plurality of thermal loads via respective control valves, including a first thermal load and a second thermal load;

wherein the controller is configured to evaluate a criterion corresponding to whether the flash potential is sufficient to meet a predicted demand of the plurality of loads;

wherein, responsive to an outcome of the evaluation, the controller is configured to evaluate priority data specifying relative priorities of the thermal loads, and to operate in a priority discharge mode in which it controls selective discharge of steam to meet a thermal energy demand of a relatively high priority thermal load in preference to controlling selective discharge of steam to meet a thermal energy demand of a relatively lower priority thermal load.

44. An installation according to claim 27, configured to provide water in the form of steam discharged from the pressure vessel to the thermal load in an open arrangement, by which the thermal load consumes the water provided as steam without corresponding return of the water as condensate; and

wherein the controller is configured to discharge steam to the thermal load at a minimum discharge pressure of between 0.5 MPa and 1.0 MPa, or the thermal load is

configured to receive discharged steam at a minimum discharge pressure of between 0.5 MPa and 1.0 MPa.

45. An installation according to claim 27, configured to supply steam discharged from the pressure vessel to at least two thermal loads; wherein:

the thermal storage installation is configured to provide water in the form of steam discharged from the pressure vessel to one of the thermal loads in an open arrangement, by which the water is placed into contact with a foreign process fluid or article, and/or is discharged from the thermal load without corresponding return of the water as condensate; and

the thermal storage installation is configured to provide water in the form of steam discharged from the pressure vessel to one of the thermal loads in a closed loop, by which the water is at least partly returned to the pressure vessel as subcooled water, for example via a subcooled supply vessel.

46. An installation according to claim 27, wherein the pressure vessel is provided with a deaerator configured to receive an inlet flow of subcooled water along a deaeration path, and configured to direct a deaerating flow of steam from within the pressure vessel in counterflow along the deaeration path;

wherein the controller is configured to vary a velocity of the deaerating flow of steam as a function of a temperature of the inlet flow and/or a temperature of the steam, to target the inlet flow reaching a saturation temperature corresponding to the steam along the deaeration path, over a range of storage pressures.

47. An installation according to claim 27, wherein the heater is installed within the pressure vessel so that a lower limit liquid level for operation of the heater corresponds to a liquid fraction of water within the pressure vessel of no more than 60%.

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